

Chapter 10

Overlapping Generations

This chapter describes the pure exchange overlapping generations model of Paul Samuelson (1958). We begin with an abstract presentation that treats the overlapping generations model as a special case of the chapter 9 general equilibrium model with complete markets and all trades occurring at time 0. There is a special type of heterogeneity across agents: each person cares about consumption only at two adjacent dates, and the set of individuals who care about consumption at a particular date includes some who care about consumption one period earlier and others who care about consumption one period later. We shall study how this special preference and demographic pattern affects some outcomes of the chapter 9 model.

Complete markets with time 0 trading in an overlapping generations model strains credulity by imagining that equilibrium price and quantity sequences are set at time 0, before most of the people who are to execute trades have been born. For that reason, most applied work with the overlapping generations model assumes a sequential-trading arrangement, like the sequential trade in Arrow securities described in chapter 9. In a sequential-trading arrangement, all trades are agreed to only by living agents.

Despite its implausibility, we use the abstraction of complete markets with time 0 trading to help us understand properties of overlapping generations models that include possible multiplicity of equilibria and Pareto inefficient equilibrium allocations, outcomes that don't happen in the chapter 9 general equilibrium model. Since we assume that everyone can trade with anyone else at time 0, we'll discover that such outcomes don't hinge on agents alive today not being able to trade with agents not yet alive.¹

Later in this chapter, we use versions of the model with sequential trading to tell how the overlapping generations model provides a framework for thinking about equilibria with valued fiat currency, government debt, intergenerational transfers, and fiscal policy. **We conclude with the analysis of a growth model.**

¹ In section 10.3, under time 0 trading, we will describe an alternative equilibrium concept that entails a clearinghouse.

10.1. Preferences and endowments

Time is discrete, starts at $t = 0$, and lasts forever, so $t = 0, 1, 2, \dots$. In each period, there is a large number N of identical agents born who live for two periods. The assumption of a large cohort N justifies an analysis of a competitive equilibrium in which agents take equilibrium prices as given. For expositional simplicity we assume that there exists a representative agent in each generation. There is an infinity of representative agents, one for each generation, named $i = -1, 0, 1, \dots$. We can also regard i as agent i 's period of birth.

There is a single good at each date. The good is not storable. There is no uncertainty. Each agent has a strictly concave, twice continuously differentiable, one-period utility function $u(c)$ that is strictly increasing in consumption c of the one good. Agent i orders consumption stream $\mathbf{c}^i = \{c_t^i\}_{t=0}^\infty$ according to the special utility functional

$$U^i(\mathbf{c}^i) = u(c_i^i) + u(c_{i+1}^i), \quad i \geq 0, \quad (10.1.1a)$$

$$U^{-1}(\mathbf{c}^{-1}) = u(c_0^{-1}). \quad (10.1.1b)$$

As a superscript, i denotes a person's birth date; as a subscript, it denotes the current date. Notice that agent $i \geq 0$ only wants goods dated i and $i + 1$, while an agent born at $i = -1$, before the model starts at date $i = 0$, wants consumption only at time 0. Equations (10.1.1) tell us that agents want to consume only when they are alive in periods $t \geq 0$.

Each consumer $i \geq -1$ has an endowment sequence $\mathbf{y}^i = \{y_t^i\}_{t=0}^\infty$.

DEFINITIONS: An *allocation* is a list $\{\mathbf{c}^i\}_{i \geq -1} = \{\{c_t^i\}_{t \geq 0}\}_{i \geq -1}$ of each agent i 's consumption at every date t . A *feasible allocation* satisfies

$$\sum_i c_t^i \leq \sum_i y_t^i \quad (10.1.2)$$

for all $t \geq 0$.

In most of our analysis, we will assume that consumers are endowed with goods only when they are alive. That is, each consumer $i \geq 0$ has an endowment sequence $\mathbf{y}^i$ satisfying $y_i^i \geq 0$, $y_{i+1}^i \geq 0$, $y_t^i = 0 \forall t \neq i \text{ or } i + 1$; whereas the initial old consumer $i = -1$ has a single endowment $y_0^{-1} \geq 0$. Evidently an allocation is then feasible if for all $t \geq 0$,

$$c_t^t + c_t^{t-1} \leq y_t^t + y_t^{t-1}. \quad (10.1.3)$$

Furthermore, we will mostly focus on economies with stationary endowments in which $y_i^i = y_1$ and $y_i^{i-1} = y_2$ for all $i \geq 0$. Here subscript 1 on y_1 denotes young and 2 on y_2 denotes old.

10.2. Dynamic (in)efficiency

The Pareto inefficient equilibria encountered in this chapter reflect intertemporal inefficiencies. Such equilibria are commonly called *dynamically inefficient*. Under the preference specification U^i in (10.1.1) and stationary endowments, $(y_i^i, y_i^{i-1}) = (y_1, y_2)$ for all $i \geq 0$ and $y_i^i = 0$ otherwise, multiple equilibria and dynamically inefficient equilibria both exist if and only if $y_2 < y_1$. If $y_2 \geq y_1$, there is a unique equilibrium that is also Pareto efficient, as we found more generally for the chapter 9 general equilibrium model.

10.3. Time 0 trading

As in chapter 9, we begin by analyzing an economy in which all trades take place at time 0. But instead of a “Walrasian auctioneer” posting equilibrium prices we assume that there is a “clearinghouse” that posts prices, monitors demands and supplies for goods in all periods, and verifies that each person’s single time 0 budget constraint is satisfied. An equilibrium price vector assures that for periods $t \geq 1$ aggregate endowments in that period are consumed by living agents, but it can turn out that in period 0 the aggregate endowment exceeds aggregate consumption which would mean that the clearinghouse ends up with some time 0 good not allocated. If that happens we simply assume that the excess supply of the time 0 good, denoted $\Omega_0 \geq 0$, is left to perish because no agents who want those goods can also afford to buy them. Hence, the market clearing conditions under the alternative equilibrium concept with a clearinghouse are

$$c_t^t + c_t^{t-1} = y_t^t + y_t^{t-1}, \quad \forall t \geq 1; \quad (10.3.1a)$$

$$c_0^0 + c_0^{-1} + \Omega_0 = y_0^0 + y_0^{-1}, \quad (10.3.1b)$$

where $\Omega_0 \geq 0$.

Excess supply of time 0 goods cannot occur in the chapter 9 general equilibrium model. Consequently, it doesn't matter whether we compute a Walrasian equilibrium or instead use our alternative equilibrium concept that entails a clearinghouse for that economy. Both equilibrium concepts lead to the same unique equilibrium allocation that we characterized in chapter 9. "Walras' Law" holds in that economy. Restated in terms of our present setup, Walras' Law would assert that if all time 0 markets for goods dated $t \geq 1$ clear, then so must the remaining market $t = 0$. But we can and shall show that Walras' Law can be violated in an overlapping generations model. So here it matters that we adopt our alternative equilibrium concept with its clearinghouse; this equilibrium concept lets us analyze a richer set of possible outcomes than encompassed within a Walrasian equilibrium. Indeed, inspired by the additional outcomes that can emerge under the alternative equilibrium concept with a clearinghouse, in later sections of this chapter we shall study sequential trading equilibria in which fiat money and unbacked government debt are valued in equilibrium, outcomes that are not possible in the chapter 9 general equilibrium model.

10.3.1. Agents' optimization

Thus, at time 0, there are complete markets in time $t \geq 0$ consumption goods with prices $\{q_t^0\}_{t \geq 0}$.² Agent i 's budget constraint is

$$\sum_{t=0}^{\infty} q_t^0 c_t^i \leq \sum_{t=0}^{\infty} q_t^0 y_t^i. \quad (10.3.2)$$

Letting μ^i be a Lagrange multiplier attached to the budget constraint of consumer $i \geq 0$, the consumer's first-order conditions are

$$\mu^i q_i^0 = u'(c_i^i), \quad (10.3.3a)$$

$$\mu^i q_{i+1}^0 = u'(c_{i+1}^i), \quad (10.3.3b)$$

$$c_t^i = 0 \text{ if } t \notin \{i, i+1\}. \quad (10.3.3c)$$

² As shown in chapter 9, only relative prices are determined in an equilibrium. For absolute prices, we need to choose a numeraire. Here we have chosen the good at time 0 as the numeraire. Hence, using our notation in chapter 9, q_t^0 is the price of a unit of the good to be delivered at time t in terms of units of the good at time 0.

The consumer's Euler equation can be deduced from equations (10.3.3a) and (10.3.3b),

$$\frac{u'(c_{i+1}^i)}{u'(c_i^i)} = \frac{q_{i+1}^0}{q_i^0} \equiv \alpha_i = \frac{1}{R_i}, \quad (10.3.4)$$

where α_i is the price of one good dated $i + 1$ in terms of the good dated i .³ Later, in a sequential trading equilibrium, α_i will be the inverse of the one-period gross interest rate R_i between periods i and $i + 1$.

As for an initial old agent $i = -1$ who is alive only in the first period of the economy, the optimal consumption choice is simply to consume his entire wealth in period 0. Under our specialization of endowment streams, the initial old consumer's budget constraint is $q_0^0 c_0^{-1} \leq q_0^0 y_0^{-1}$ and hence, the initial old agent's optimal consumption choice is $c_0^{-1} = y_0^{-1}$.

10.3.2. Example equilibria

Let $\epsilon \in (0, .5)$. Endowments are

$$\begin{aligned} y_i^i &= 1 - \epsilon, \quad \forall i \geq 0, \\ y_{i+1}^i &= \epsilon, \quad \forall i \geq -1, \\ y_t^i &= 0 \text{ otherwise.} \end{aligned} \quad (10.3.5)$$

This economy has many equilibria. We describe two stationary equilibria now and nonstationary equilibria later. An equilibrium is said to be a stationary equilibrium if its allocation is stationary in the following sense.

DEFINITION: An allocation is *stationary* if $c_i^i = c_1$ and $c_{i+1}^i = c_2$ for all $i \geq 0$.

Here the subscript 1 on c_1 denotes young and 2 on c_2 denotes old. Note that we do not require that $c_0^{-1} = c_2$. Indeed, as explained above, equilibrium consumption of the initial old agent $i = -1$ will be $c_0^{-1} = y_0^{-1} = \epsilon$ in both stationary and nonstationary equilibria.

We can use a guess-and-verify method to confirm two stationary equilibria. Specifically, we proceed by first guessing the equilibrium allocation for agents $i \geq 0$, which amounts to guessing c_1 and c_2 in a stationary equilibrium. Given that guess, we use Euler equation (10.3.4) to deduce at which relative price q_{i+1}^0/q_i^0

³ If we had instead used our chapter 9, α_t would have been written as q_{t+1}^t .

such a consumption profile (c_1, c_2) is consistent with utility maximization by agent $i \geq 0$. Next, given that deduced relative price and the guess of an allocation, we check that budget constraint (10.3.2) holds with equality for each agent $i \geq 0$, which here becomes

$$c_1 + \frac{q_{i+1}^0}{q_i^0} c_2 = (1 - \epsilon) + \frac{q_{i+1}^0}{q_i^0} \epsilon, \quad (10.3.6)$$

where we have divided both sides of the budget constraint by q_i^0 and hence, changed the numeraire to be the good dated i . Finally, we verify that markets clear in accordance with equations (10.3.1).

So by using this guess-and-verify method, we can confirm the following two stationary equilibria.

1. **Equilibrium H:** a high-interest-rate equilibrium. We guess a stationary equilibrium allocation $c_1 = c_2 = .5$ and $c_0^{-1} = \epsilon$. To be consistent with utility maximization, Euler equation (10.3.4) must be satisfied and hence, the relative price would have to be $q_{i+1}^0/q_i^0 = u'(.5)/u'(.5) = 1 = \alpha_H$. Given our normalization $q_0^0 = 1$, the sequence of absolute prices becomes $q_t^0 = 1$ for all $t \geq 0$. We can then verify that budget constraint (10.3.6) holds with equality for each agent $i \geq 0$, and trivially for the initial old agent $i = -1$ who simply consumes his endowment in period 0, $c_0^{-1} = \epsilon$. Concerning market clearing conditions, equation (10.3.1a) holds since aggregate consumption equals aggregate endowments in periods $t \geq 1$, while equation (10.3.1b) yields excess supply of goods dated 0, namely,

$$\Omega_0 = y_0^0 + y_0^{-1} - c_0^0 - c_0^{-1} = .5 - \epsilon > 0,$$

where the quantity $\Omega_0 > 0$ of goods dated 0 is left to perish since no agent who wants those goods can afford to buy them.

2. **Equilibrium L:** a low-interest-rate equilibrium. We guess a stationary equilibrium allocation $c_1 = 1 - \epsilon$, $c_2 = \epsilon$, and $c_0^{-1} = \epsilon$. For that consumption profile to be consistent with utility maximization, the relative price would have to be $q_{i+1}^0/q_i^0 = u'(\epsilon)/u'(1 - \epsilon) = \alpha_L > 1$, i.e., Euler equation (10.3.4) must hold for all $i \geq 0$. Given our normalization $q_0^0 = 1$, the sequence of absolute prices becomes $q_t^0 = (\alpha_L)^t$ for all $t \geq 0$. Since all agents are simply consuming their endowments, $c_t^i = y_t^i$ for all i, t , budget constraints (10.3.6) and market clearing conditions (10.3.1) are trivially satisfied, and

there is no excess supply of goods dated 0, $\Omega_0 = 0$. This equilibrium is autarkic, with equilibrium prices eradicating all incentives to trade.

To understand why we call them high- and low-interest-rate equilibria, recall that the relative price α_t in expression (10.3.4) would be the inverse of the one-period gross interest rate R_t in a sequential trading equilibrium. Thus, the interest rates associated with the high- and low-interest-rate equilibrium are $R_H = 1$ and $R_L = (\alpha_L)^{-1} < 1$, respectively. Soon we shall construct equilibria in which R_t varies over time t .

10.3.3. Relation to welfare theorems

Equilibrium H Pareto dominates equilibrium L. In equilibrium H, every generation after the initial old one is better off and no generation is worse off than in equilibrium L. The equilibrium H allocation is peculiar because some of the time 0 good is not consumed, leaving room to set up a giveaway program to the initial old that makes them better off. We shall soon learn how an institution of either fiat money or **of unbacked, perpetual government debt** can accomplish this purpose.⁴

With equality between supply and demand in all periods, equilibrium L is also a Walrasian equilibrium. But, evidently, this competitive equilibrium fails to satisfy the assumptions needed to deliver the first fundamental theorem of welfare economics that isolates conditions under which a competitive equilibrium allocation is Pareto optimal.⁵ The condition that is violated by equilibrium L is the assumption that the value of the aggregate endowment at the equilibrium prices is finite.⁶

⁴ See Karl Shell (1971) for an investigation that characterizes why some competitive equilibria in overlapping generations models fail to be Pareto optimal. As explained by Shell, “the double infinity of traders and dated commodities allows for competitive equilibria that are not Pareto-optimal [and] the fact that generations do not meet is not essential.”

⁵ See Mas-Colell, Whinston, and Green (1995) and Debreu (1954).

⁶ Note that if the horizon of the economy were finite, then the counterpart of equilibrium H would not exist and the allocation of the counterpart of equilibrium L would be Pareto optimal.

10.3.4. Nonstationary equilibria

Our example economy has more equilibria. To construct them, we use an offer curve to summarize preferences and optimal consumption decisions. We describe a graphical apparatus proposed by David Gale (1973) and used to great advantage by William Brock (1990).

DEFINITION: Agent i 's *offer curve* is a locus of (c_i^i, c_{i+1}^i) that solves

$$\max_{\{c_i^i, c_{i+1}^i\}} U^i(\mathbf{c}^i)$$

subject to

$$c_i^i + \alpha_i c_{i+1}^i \leq y_i^i + \alpha_i y_{i+1}^i.$$

Here $\alpha_i \equiv \frac{q_{i+1}^0}{q_i^0}$, the reciprocal of the one-period gross rate of return from period i to $i+1$, is a parameter that we shall use to sweep out a (c_i^i, c_{i+1}^i) locus.

Evidently, points (c_i^i, c_{i+1}^i) on the offer curve simultaneously solve the following pair of equations:

$$c_i^i + \alpha_i c_{i+1}^i = y_i^i + \alpha_i y_{i+1}^i \quad (10.3.7a)$$

$$\frac{u'(c_{i+1}^i)}{u'(c_i^i)} = \alpha_i \quad (10.3.7b)$$

for different values of the relative price $\alpha_i > 0$. We denote the offer curve swept out by $\alpha_i > 0$ by

$$\psi(c_i^i, c_{i+1}^i) = 0.$$

The offer curve is illustrated in Figure 10.3.1. We trace it out by varying α_i in the consumer's problem and reading off tangency points between the consumer's indifference curve and the budget line. So at each point on the offer curve, there is an indifference curve whose tangent at that point is a straight line that can be extended to go through the consumer's endowment point. Hence, the locus of the offer curve depends on the endowment vector and lies above the indifference curve through the endowment point. Since the consumer can always choose to consume his endowment, an optimally chosen consumption (c_i^i, c_{i+1}^i) could never lie on an indifference curve lower than the indifference curve through the endowment point.

By relabeling axes, in the same graph we can plot a consumption possibility frontier

$$c_i^i + c_i^{i-1} \leq y_i^i + y_i^{i-1} \quad (10.3.8)$$

that bounds feasible allocations at time i , which we call the *feasibility line*. Note that while the offer curve restricts a life-time consumption profile (c_i^i, c_{i+1}^i) for one person born at time i , the feasibility line involves pairs (c_i^i, c_i^{i-1}) of consumptions assigned to a young and old person at time i .⁷

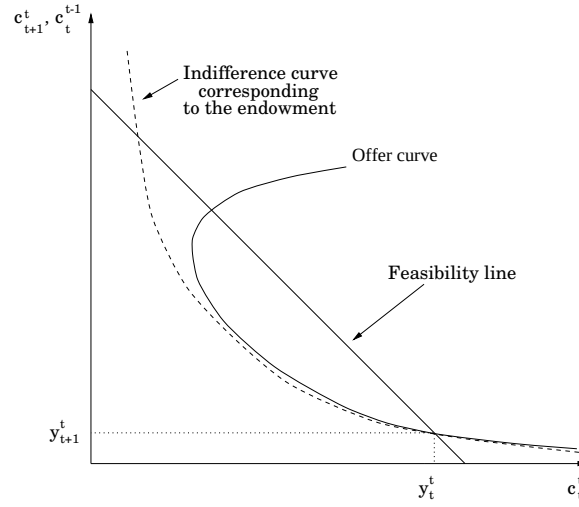


Figure 10.3.1: The offer curve and feasibility line.

Following Gale (1973), for the important special case in which $(y_i^i, y_i^{i-1}) = (y_1, y_2)$ for all $i \geq 0$, we can use the offer curve and a straight line depicting feasibility in the (c_i^i, c_i^{i-1}) plane to compute equilibrium allocations and prices. The assumption of stationary endowments means that the feasibility line is the same in all periods $i \geq 0$. As explained earlier, the initial old agent consumes his endowment in an equilibrium, $c_0^{-1} = y_2$. As for the continuation of the

⁷ Given our assumptions on preferences and endowments, the conscientious reader will note that Figure 10.3.1 appears distorted because the offer curve really ought to intersect the feasibility line along the 45 degree line with $c_t^t = c_{t+1}^t$, i.e., at the allocation affiliated with equilibrium H above.

equilibrium allocation, there is a multiplicity of possible outcomes since $y_2 < y_1$. Specifically, we can initiate an equilibrium computation by choosing $c_0^0 \in ((y_1 + y_2)/2, y_1)$.⁸ As we shall see, for each different c_0^0 , a different competitive equilibrium prevails. Thus, we can find an equilibrium allocation by solving a pair of nonlinear difference equations. For $i \geq 0$, the equations are

$$\psi(c_i^i, c_{i+1}^i) = 0, \quad (10.3.9a)$$

$$c_{i+1}^{i+1} + c_{i+1}^i = y_1 + y_2. \quad (10.3.9b)$$

Equation (10.3.9a) states that the allocation to an agent born at time i is on the offer curve. Equation (10.3.9b) asserts that the allocation at time $i + 1$ is feasible.⁹ So we take c_0^0 as an initial condition and then compute an equilibrium allocation via the following steps:

- (a) Solve agent 0's offer curve (10.3.9a) for c_1^0 .
- (b) Solve feasibility condition (10.3.9b) for c_1^1 .
- (c) Iterate on these steps indefinitely and hope for convergence.

After an allocation has been computed, a time 0 trading equilibrium price system can be computed from

$$q_{i+1}^0 = q_i^0 \frac{u'(c_{i+1}^i)}{u'(c_i^i)}$$

for all $i \geq 0$, given the numeraire $q_0^0 = 1$.

⁸ By setting c_0^0 equal to one of the two boundary points, $(y_1 + y_2)/2$ or y_1 , we could recover one of the two section 10.3.2 stationary equilibria.

⁹ Feasibility in period $i = 0$ is assured by our choice of $c_0^0 < y_1$ and the initial old agent consuming his endowment in any equilibrium, $c_0^{-1} = y_2$, i.e., equation (10.3.1b) is satisfied with $\Omega_0 > 0$.

10.3.5. Computing equilibria

We now describe David Gale's (1973) equilibrium computation machine that consists of a graph like that illustrated in Figure 10.3.2. Notice that y_t^t and c_t^t are on the horizontal axis while y_{t+1}^t and both c_{t+1}^t and c_t^{t-1} are on the vertical axis. The straight feasibility line with slope -1 goes through endowment point (y_t^t, y_{t+1}^t) and plots feasible pairs (c_t^t, c_t^{t-1}) that can be allocated to people alive at time t . The curved offer curve also goes through the endowment point (y_t^t, y_{t+1}^t) and plots pairs (c_t^t, c_{t+1}^t) that a person born at t optimally chooses as a function of the relative price α_t implicitly sweeping out the offer curve.

Applying the section 10.3.4 algorithm, we start with a proposed c_0^0 , a time 0 allocation to the initial young. The candidate time 0 equilibrium allocation is feasible as explained in footnote 9, but the time 0 young will choose c_0^0 only if the price α_0 is such that (c_0^0, c_1^0) lies on the offer curve. Therefore, we choose c_1^0 from the point on the offer curve that cuts a vertical line through c_0^0 . Then we proceed to find c_1^1 from the intersection of a horizontal line through c_1^0 and the feasibility line. We continue recursively in this way, choosing c_i^i at the intersection of the feasibility line with a horizontal line through c_{i-1}^{i-1} , then choosing c_{i+1}^i at the intersection of a vertical line through c_i^i and the offer curve. We can construct a sequence of α_i 's from the slope of a straight line through the endowment point and the sequence of (c_i^i, c_{i+1}^i) pairs that lie on the offer curve.

If the offer curve has the shape drawn in Figure 10.3.2, any c_0^0 between the two intersections of the offer curve and the feasibility line is an equilibrium setting of c_0^0 . Each such c_0^0 is associated with a distinct allocation $\{c_i^i, c_{i+1}^i\}_{i \geq 0}$ and price system $\{\alpha_i\}_{i \geq 0}$, while consumption of the initial old agent is the same across all equilibria, $c_0^{-1} = y_2$. If c_0^0 is chosen to be the lower boundary point for that set, the economy will simply remain in that *high*-interest-rate stationary equilibrium. For all other choices of c_0^0 , the economy converges to the *low*-interest-rate stationary equilibrium allocation and interest rate.

Log utility: Suppose that $u(c) = \ln(c)$ and that the endowment is described by equations (10.3.5) with $\epsilon \in (0, .5)$. After solving equations (10.3.7) for that specification, the offer curve is given by the recursions

$$c_i^i = .5(1 - \epsilon + \alpha_i \epsilon), \quad (10.3.10a)$$

$$c_{i+1}^i = \frac{1}{\alpha_i} c_i^i. \quad (10.3.10b)$$

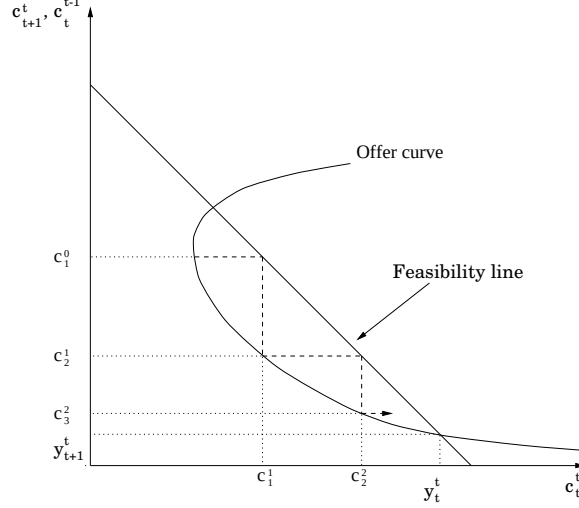


Figure 10.3.2: A nonstationary equilibrium allocation.

Hence, feasibility at $i \geq 1$ and the offer curves then imply

$$\frac{1}{2}(1 - \epsilon + \alpha_i \epsilon) + \frac{1}{2\alpha_{i-1}}(1 - \epsilon + \alpha_{i-1} \epsilon) = 1. \quad (10.3.11)$$

This implies the difference equation

$$\alpha_i = \epsilon^{-1} - \frac{\epsilon^{-1} - 1}{\alpha_{i-1}}. \quad (10.3.12)$$

To find the stationary equilibria of this economy, we let $\alpha_i = \alpha$ for all i in difference equation (10.3.12) and after rearranging,

$$\alpha^2 - \epsilon^{-1}\alpha + (\epsilon^{-1} - 1) = 0. \quad (10.3.13)$$

Using the quadratic formula, the equilibrium prices in two stationary equilibria are:

$$\alpha = \frac{\epsilon^{-1} \pm \sqrt{\epsilon^{-2} - 4(\epsilon^{-1} - 1)}}{2} = \frac{\epsilon^{-1} \pm \sqrt{(\epsilon^{-1} - 2)^2}}{2} = \begin{cases} 1, \\ \frac{1 - \epsilon}{\epsilon} \end{cases} \quad (10.3.14)$$

The two equilibria are the log-utility versions of the two stationary equilibria in section 10.3.2, $\alpha_H = 1$ and $\alpha_L = u'(\epsilon)/u'(1 - \epsilon) = (1 - \epsilon)/\epsilon > 1$. To initiate the

computation of a nonstationary equilibrium using difference equation (10.3.12), we choose $\alpha_0 \in (\alpha_H, \alpha_L)$, which is tantamount to choosing $c_0^0 \in (.5, 1 - \epsilon)$ when using the section 10.3.4 algorithm. The limiting value of $\{\alpha_i\}$ for any nonstationary equilibrium is α_L . Hence, the autarkic equilibrium α_L is stable, while the stationary equilibrium $\alpha_H = 1$ is not.

10.4. Sequential trading

We now alter the trading arrangement to bring it into line with more common presentations of the overlapping generations model. We abandon the time 0, complete markets trading arrangement and replace it with sequential trading. In our model with two-period lived agents, the only equilibrium is then autarky since there is no double coincidence of wants. While a young agent might want to buy or sell goods in exchange for goods next period, a current old agent will not be around next period. Hence, the unique equilibrium price system eradicates all incentives to trade, $\alpha_t = u'(y_{t+1}^t)/u'(y_t^t)$ for all $t \geq 0$.

To create opportunities for trade across generations, we introduce a government that at time 0 issues a zero-dividend infinitely durable asset in the form of fiat money or issues one-period bonds that are rolled over forever, i.e., the government debt is unbacked. Under the auxiliary assumption that the government gives this asset to the initial old agent, we will show that all the equilibria of the time 0 trading economy can reemerge as sequential trading equilibria, but with a difference. Any goods that the clearinghouse had earlier left to perish at time 0, $\Omega_0 \geq 0$, will no longer be wasted but be consumed by the initial old agent. We study sequential trading equilibria under the common notion of a Walrasian auctioneer who announces equilibrium prices at which the goods and asset markets clear in every period.

The assumption that the government-issued asset is given to the initial old generation is critical for delivering equivalence of outcomes under time 0 trading and sequential trading with respect to equilibrium prices and the allocation $\{c_t^t, c_{t+1}^t\}_{t \geq 0}$ (while leaving aside the value of c_0^{-1}). Any giveaway program to the initial old generation has no effects on the equilibrium price system, since those old agents optimally consume all their wealth in period 0. If the assets

had instead been given to the initial young generation, it would have changed their offer curves and hence, affected equilibrium prices.

10.4.1. Government budget constraint

In the following sections, we will activate various items in the following government budget constraints:

$$\frac{M_t - M_{t-1}}{p_t} + \tau_t^{t-1} + \tau_t^t + \frac{B_{t+1}}{R_t} = g_t + B_t, \quad \forall t \geq 0. \quad (10.4.1)$$

Here M_t is the nominal quantity of money at the end of period t , p_t is the price level in period t , τ_t^i is a lump-sum tax levied in period t on the representative agent born in period $i \in \{t-1, t\}$ (a negative tax is a lump-sum transfer), B_{t+1} is one-period real bonds issued by the government in period t which pay B_{t+1} units of goods upon maturity in period $t+1$, R_t is the gross real one-period return on those bonds, and g_t is government purchases of goods at time t . Thus, the left side of budget constraint (10.4.1) sums sources of government funds – real seigniorage from money creation, tax revenues, and the issue of government bonds – which are used to pay for goods purchases and the maturing debt on the right side of (10.4.1). The economy starts at time $t = 0$ from initial conditions $M_{-1} = B_0 = 0$. We assume that government purchases $\{g_t\}_{t \geq 0}$ provide no utility to agents.

10.5. Fiat money

In Samuelson's (1958) model, trading occurs sequentially through a medium of exchange, an inconvertible (or "fiat") currency. Preferences and endowments are as described above, with one important additional component of the endowment. At date $t = 0$, old agents are endowed in the aggregate with $M > 0$ units of paper currency that no one has promised to redeem for goods. This paper currency is not "backed" by another asset. Nevertheless, Samuelson showed that there exists a system of expectations about the value of paper currency that can impart positive value to an unbacked currency. Currency will be valued today if people expect it to be valued tomorrow. In this way Samuelson constructed a situation in which currency is backed by expectations without promises.

To cast this Samuelson (1958) setup in terms of our government formulation in (10.4.1), we assume that the government at time 0 gives initial old agents a total of M units of currency. Consequently, the government budget constraint at time 0 becomes

$$\frac{M_0}{p_0} + \tau_0^{-1} = 0, \quad (10.5.1)$$

where $M_0 = M$. The real value of the lump-sum transfer of M that the government hands to initial old agents depends on the equilibrium price level $p_0 > 0$, so that in effect $-\tau_0^{-1} = M/p_0 \geq 0$. Thereafter, the government takes no further actions.

At each date $t \geq 0$, a young agent purchases m_t^t units of currency at a price of $1/p_t$ units of the time t consumption good. Here $p_t > 0$ is the time t price level. At each $t \geq 0$, each old agent exchanges his holdings of currency for the time t consumption good. The budget constraints of a young agent born in period $t \geq 0$ are

$$c_t^t + \frac{m_t^t}{p_t} \leq y_t^t, \quad (10.5.2)$$

$$c_{t+1}^t \leq y_{t+1}^t + \frac{m_t^t}{p_{t+1}}, \quad (10.5.3)$$

$$m_t^t \geq 0. \quad (10.5.4)$$

If money is valued and, hence, held by agents, inequalities (10.5.2) and (10.5.3) imply

$$c_t^t + c_{t+1}^t \left(\frac{p_{t+1}}{p_t} \right) \leq y_t^t + y_{t+1}^t \left(\frac{p_{t+1}}{p_t} \right). \quad (10.5.5)$$

Provided that we set

$$\frac{p_{t+1}}{p_t} = \alpha_t = \frac{q_{t+1}^0}{q_t^0},$$

this budget set is identical with the one implied by equation (10.3.2) for agent $t \geq 0$. However, the budget set for an initial old agent $t = -1$ differs from that implied by equation (10.3.2) and is now given by

$$c_0^{-1} \leq y_0^{-1} - \tau_0^{-1}. \quad (10.5.6)$$

where $-\tau_0^{-1} = M/p_0 > 0$ by the government's budget constraint (10.5.1).

We use the following definitions:

DEFINITION: A nominal price sequence is a strictly positive sequence $\{p_t\}_{t \geq 0}$.

DEFINITION: An equilibrium with valued fiat money is a feasible allocation and a nominal price sequence with $p_t < +\infty$ for all $t \geq 0$ such that given the price sequence, the allocation solves the agent's problem for each generation $t \geq -1$.

The qualification that $p_t < +\infty$ for all t means that fiat money is valued. We call an equilibrium with valued fiat money a 'monetary equilibrium'. If $p_t = +\infty$, we call it a 'nonmonetary equilibrium'.

10.5.1. Computing equilibria with valued fiat currency

Summarize a consumer's optimal decisions with a saving function

$$y_t^t - c_t^t = s(\alpha_t; y_t^t, y_{t+1}^t). \quad (10.5.7)$$

Then the equilibrium conditions are

$$\frac{M}{p_t} = s(\alpha_t; y_t^t, y_{t+1}^t), \quad (10.5.8a)$$

$$\alpha_t = \frac{p_{t+1}}{p_t}, \quad (10.5.8b)$$

where it is understood that $c_{t+1}^t = y_{t+1}^t + \frac{M}{p_{t+1}}$. Equation (10.5.8a) states that at time t net saving of generation t (the expression on the right side) equals net dissaving of generation $t-1$ (the expression on the left side). To compute an equilibrium, we solve the difference equations (10.5.8) for $\{p_t\}_{t \geq 0}$, then get the allocation from the consumer's budget constraints evaluated at equality at an equilibrium level of real balances M/p_t .

As an example, suppose that $u(c) = \ln(c)$, and that $(y_t^t, y_{t+1}^t) = (w_1, w_2)$ with $w_1 > w_2$. The saving function is $s(\alpha_t; w_1, w_2) = .5(w_1 - \alpha_t w_2)$. Then equation (10.5.8a) becomes

$$.5(w_1 - w_2 \frac{p_{t+1}}{p_t}) = \frac{M}{p_t}$$

or

$$p_t = 2M/w_1 + \left(\frac{w_2}{w_1}\right) p_{t+1}. \quad (10.5.9)$$

This is a difference equation whose solutions with positive price levels are

$$p_t = \frac{2M}{w_1(1 - \frac{w_2}{w_1})} + \kappa \left(\frac{w_1}{w_2} \right)^t, \quad (10.5.10)$$

for any scalar $\kappa \geq 0$.¹⁰ The solution $\kappa = 0$ is the unique stationary solution. Solutions with $\kappa > 0$ have uniformly higher price levels than the $\kappa = 0$ solution, and have values of currency approaching zero in the limit as $t \rightarrow +\infty$.

It is instructive to compare the multiple monetary equilibria implied by equation (10.5.10), indexed by $\kappa \in [0, \infty)$, to the time 0 trading equilibria computed in section 10.3.5. The stationary monetary equilibrium with a constant price level, $\kappa = 0$, is the counterpart to the high-interest-rate stationary equilibrium under time 0 trading, α_H . Since the two stationary equilibria share the same price of one good dated $t+1$ in terms of goods dated t , $p_{t+1}/p_t = 1 = \alpha_H$, the consumption choices of agents born in period $t \geq 0$ will also be the same. In particular, given $\kappa = 0$, the constant real money supply implied by equation (10.5.10) is $M/p_t = (w_1 - w_2)/2$ for all $t \geq 0$ and, hence, the consumption allocation in the stationary monetary equilibrium of the representative agent born in period $t \geq 0$ is $c_t^t = c_{t+1}^t = (w_1 + w_2)/2$. This perfect consumption smoothing corresponds to $c_t^t = c_{t+1}^t = .5$ for $t \geq 0$ in the high-interest-rate stationary equilibrium under time 0 trading, given our earlier parameterization $y_t^t = 1 - \epsilon$ and $y_{t+1}^t = \epsilon \in (0, .5)$. (Though, note that the monetary equilibrium and the time 0 trading equilibrium differ with respect to the consumption of the initial old agents in period 0, as we discussed above and will revisit below.)

Another similarity of the stationary monetary equilibrium and the high-interest-rate stationary equilibrium under time 0 trading is that these equilibria are not stable. In fact, all other equilibria converge to autarky, i.e., the low-interest-rate stationary equilibrium under time 0 trading, α_L . To confirm that the relative price p_{t+1}/p_t in all other monetary equilibria, $\kappa \in (0, \infty)$, converges to $\alpha_L = y_t^t/y_{t+1}^t$ as derived in section 10.3.5, we compute

$$\lim_{t \rightarrow \infty} \frac{p_{t+1}}{p_t} \Big|_{\kappa \in (0, \infty)} = \lim_{t \rightarrow \infty} \frac{\frac{2M}{w_1(1 - \frac{w_2}{w_1})} + \kappa \left(\frac{w_1}{w_2} \right)^{t+1}}{\frac{2M}{w_1(1 - \frac{w_2}{w_1})} + \kappa \left(\frac{w_1}{w_2} \right)^t} \Big|_{\kappa \in (0, \infty)} = \frac{w_1}{w_2}.$$

¹⁰ See appendix B from chapter 2.

10.5.2. Equivalence of equilibria

The similarities across equilibria discussed in the previous section motivate us to formulate a proposition that describes how, after an adjustment to the endowment and the consumption allocated to the initial old, a competitive equilibrium allocation with time 0 trading is a competitive equilibrium allocation in the fiat money economy with sequential trading.

PROPOSITION: Let $\bar{c}^i$ denote a competitive equilibrium allocation with time 0 trading and suppose that it satisfies $\bar{c}_0^0 < y_0^0$. There exists an equilibrium allocation of the monetary economy with sequential trading that satisfies $c_i^i = \bar{c}_i^i$, $c_{i+1}^i = \bar{c}_{i+1}^i$ for $i \geq 0$.

PROOF: Take the competitive equilibrium allocation and price system under time 0 trading and form $\alpha_i = q_{i+1}^0/q_i^0$. Set $m_i^i/p_i = y_i^i - \bar{c}_i^i$. Set $m_i^i = M$ for all $i \geq 0$, and determine p_0 from $\frac{M}{p_0} = y_0^0 - \bar{c}_0^0$. This last equation determines a positive initial price level p_0 provided that $y_0^0 - \bar{c}_0^0 > 0$. Determine subsequent price levels from $p_{i+1} = \alpha_i p_i$. Assign an allocation to the initial old: $c_0^{-1} = y_0^{-1} + \frac{M}{p_0} = y_0^{-1} + (y_0^0 - \bar{c}_0^0)$. ■

In a monetary equilibrium, time t real balances equal both the per capita *saving* of the young and the per capita *dissaving* of the old. To be a monetary equilibrium, both quantities must be positive for all $t \geq 0$.

A converse of the proposition is true.

PROPOSITION: Let $\bar{c}^i$ be an equilibrium allocation for the fiat money economy. Then there is a competitive equilibrium with time 0 trading with the same allocation, provided that the endowment of the initial old is augmented with an appropriate transfer from the clearinghouse.

To verify this proposition, we have to construct the required transfer from the clearinghouse to the initial old. Evidently, it is $y_0^0 - \bar{c}_0^0$. We invite the reader to complete the proof.

10.6. Unbacked government debt

We now study how to transform the fiat monetary equilibria of the preceding section into equilibria with unbacked government debt. Instead of emitting fiat currency in period 0, the government finances the lump-sum transfer $-\tau_0^{-1} > 0$ to the initial old agents in period 0 by issuing real (indexed) bonds. Specifically, let B_t be the quantity of one-period bonds issued by the government in period $t - 1$, which promise to pay B_t units of goods upon maturing in period t . The price of those bonds in period $t - 1$ is the inverse of the gross real one-period interest rate R_{t-1} . In each period $t \geq 1$, the government redeems the maturing debt by issuing new one-period bonds. Thus, the debt is unbacked since there are never any taxes levied to service the debt. The government budget constraints are

$$\tau_0^{-1} + \frac{B_1}{R_0} = 0, \quad (10.6.1a)$$

$$\frac{B_{t+1}}{R_t} = B_t, \quad \forall t \geq 1. \quad (10.6.1b)$$

We solve equations (10.6.1) forward to some date $T \geq 1$:

$$-\tau_0^{-1} = \frac{B_1}{R_0} = \frac{1}{R_0} \frac{B_2}{R_1} = \frac{1}{R_0 R_1} \frac{B_3}{R_2} = \dots = \frac{1}{\prod_{j=0}^{T-1} R_j} \frac{B_{T+1}}{R_T} \quad (10.6.2)$$

and, hence, government indebtedness in period $T \geq 1$ equals

$$\frac{B_{T+1}}{R_T} = -\tau_0^{-1} \prod_{j=0}^{T-1} R_j > 0, \quad (10.6.3)$$

which arises from the initial borrowing at time 0, $B_1/R_0 = -\tau_0^{-1}$, and compounded interest.

At each date $t \geq 0$, a young agent purchases b_{t+1}^t units of newly issued one-period government bonds at a price of $1/R_t$ units of the time t consumption good. When the bonds mature next period $t + 1$, the government pays b_{t+1}^t units of the time $t + 1$ consumption good to the agent as a bond holder. The budget constraints of a young agent born in period $t \geq 0$ are

$$c_t^t + \frac{b_{t+1}^t}{R_t} \leq y_t^t, \quad (10.6.4a)$$

$$c_{t+1}^t \leq y_{t+1}^t + b_{t+1}^t. \quad (10.6.4b)$$

Inequalities (10.6.4a) and (10.6.4b) imply

$$c_t^t + \frac{c_{t+1}^t}{R_t} \leq y_t^t + \frac{y_{t+1}^t}{R_t}. \quad (10.6.5)$$

The budget constraint of an initial old agent, $i = -1$, is

$$c_0^{-1} \leq y_0^{-1} - \tau_0^{-1}, \quad (10.6.6)$$

where $-\tau_0^{-1} = B_1/R_0 > 0$ by government budget constraint (10.6.1a).

10.6.1. Isomorphic equilibria with fiat money or unbacked debt

With a change of variables we can transform the equilibria with valued fiat money in section 10.5 into equilibria with valued unbacked government debt and vice versa. To go from a monetary equilibrium to an equilibrium with unbacked debt, set $R_t = p_t/p_{t+1}$ and $B_{t+1}/R_t = M/p_t$ for $t \geq 0$. To go in the opposite direction from an equilibrium with unbacked debt to a monetary equilibrium, pick some arbitrary nominal quantity of money M and solve for p_0 from $M/p_0 = B_1/R_0$ ($= -\tau_0^{-1}$). All other prices, $\{p_t\}_{t \geq 1}$, are then recursively set according to $p_{t-1}/p_t = R_{t-1}$ for $t \geq 1$.

To establish that each of these transformations generates an equilibrium, we must show that the resulting rates of return and amounts of assets are the same as in a stipulated original equilibrium, as well as that government budget constraints are satisfied. Note that both transformations explicitly impose same rates of return, i.e., we set R_t equal to p_t/p_{t+1} or vice versa. Furthermore, when going from a fiat-money equilibrium to an unbacked-debt equilibrium, we also set the sequences of assets to be the same, $B_{t+1}/R_t = M/p_t$ for $t \geq 0$. So it only remains to verify that those levels of government debt are consistent with government budget constraints (10.6.1). This is trivially true for constraint (10.6.1a) in period 0, given that budget constraint (10.5.1) in the stipulated original fiat-money equilibrium holds. Regarding the remaining constraints (10.6.1b), we substitute our mapping into the left side and the right side of each one of those constraints:

$$\frac{B_{t+1}}{R_t} = B_t \left(= R_{t-1} \frac{B_t}{R_{t-1}} \right) \implies \frac{M}{p_t} = \frac{p_{t-1}}{p_t} \frac{M}{p_{t-1}}, \quad \forall t \geq 1;$$

which confirms that the left side equals the right side for each one of constraints (10.6.1b) and, hence, all government budget constraints are satisfied in the unbacked-debt equilibrium. Thus, we conclude that an equilibrium with fiat money can be mapped into an equilibrium with unbacked government debt.

As for the mapping from an unbacked-debt equilibrium to a fiat-money equilibrium, besides having explicitly set rates of return equal, our described setting of p_0 ensures that the government's only budget constraint (10.5.1) is satisfied; $M/p_0 = B_1/R_0 = -\tau_0^{-1}$. So it only remains to verify that all future assets in the monetary economy are equal to those of the unbacked-debt equilibrium; that is, to verify that real money supplies, $\{M/p_t\}_{t \geq 1}$, are equal to government indebtedness in the unbacked-debt equilibrium, as given by expression (10.6.3). We confirm that to be true by substituting our mapping into the right side of expression (10.6.3) for $T \geq 1$:

$$\frac{B_{T+1}}{R_T} = -\tau_0^{-1} \prod_{j=0}^{T-1} R_j = \frac{M}{p_0} \prod_{j=0}^{T-1} \frac{p_j}{p_{j+1}} = \frac{M}{p_T}.$$

Thus, we conclude that an equilibrium with unbacked government debt can be mapped into an equilibrium with fiat money.

The isomorphic equilibria can be illustrated with our example of logarithmic preferences. As noted earlier, the multiple monetary equilibria implied by equation (10.5.10) can be indexed by $\kappa \in [0, \infty)$. The stationary monetary equilibrium, $\kappa = 0$, has a constant price level, i.e., a gross real rate of return on fiat money equal to one. When mapped into an economy with unbacked government debt, the same stationary equilibrium allocation can be attained by the government at time 0 issuing a particular amount of real bonds, let say, $B_1 = B_H$, as determined by the young generation's demand for savings at a gross real interest rate equal to $R_0 = 1$. At that interest rate, the debt can be rolled over in perpetuity while remaining constant. As we know from before, this high-interest-rate stationary equilibrium is not stable, which is also self-evident here. If the debt issued at time 0 is chosen to be lower than B_H , the net real interest rate will be negative, $R_t < 1$, and, hence, subsequent debt levels in expression (10.6.3) would be monotonically declining and converging to zero; precisely as in a monetary equilibrium indexed by $\kappa > 0$, where inflation would cause real money supplies M/p_t to decline over time and converge to zero.

While fiat-money equilibria and unbacked-debt equilibria are isomorphic, there are economically substantive differences in terms of what the government

controls in each type of equilibrium. In an unbacked-debt equilibrium, the government chooses at time 0 the quantity of indexed bonds B_1 to be issued. Furthermore, the price $1/R_0$ at which those bonds will be sold can be deduced from agents' savings functions, so the government has full control over the size of the lump-sum transfer to be handed over to the initial old agents, $-\tau_0^{-1} = B_1/R_0$. In contrast, the government in a fiat-money equilibrium chooses the amount of nominal money M to be given to the initial old agents. But recall that in our transformation of an unbacked-debt equilibrium into a fiat-money equilibrium we began by saying: 'pick some arbitrary nominal quantity of money M '; that is, the positive nominal quantity M itself bears no significance. Instead, what matters is agents' beliefs about price levels and, hence, what will be the real return on money holdings. According to equation (10.5.10), the set of possible price sequences consistent with money market clearing can be indexed by $\kappa \in [0, \infty)$. Thus, depending on what agents' beliefs are, the associated price level p_0 pins down the real value of the fiat money handed over to the initial old agents, $-\tau_0^{-1} = M/p_0$. In that sense, the government has no direct control over the real value of that transfer.¹¹

An unbacked-debt equilibrium with declining indebtedness is reminiscent of a situation where a government pays down (amortizes) its debt over time. Here, the size of that 'amortization' between period t and $t+1$ is

$$\frac{B_t}{R_{t-1}} - \frac{B_{t+1}}{R_t} = \frac{B_t}{R_{t-1}} - B_t = (1 - R_{t-1}) \frac{B_t}{R_{t-1}} \geq 0, \quad (10.6.7)$$

where we have invoked equation (10.6.1b). Next, we will examine how the 'amortization' is slowed down when the government makes unfunded purchases of goods $\{g_t\}_{t \geq 0}$, which amounts to deficit finance with unbacked debt. The analogue to such deficit finance in an isomorphic fiat-money equilibrium is that money is printed to pay for the purchases. This source of government funding is called seigniorage and it is effectively a taxation of agents' money holdings, as we will also explore below.

¹¹ A fiscal theory of the price level (FTPL) has been proposed for selecting which monetary equilibrium that will prevail (see chapters 17 and 28). In the current context, such a FTPL would point to government budget constraint (10.5.1) as the determinant of price level p_0 and, consequently, of the value of κ in equation (10.5.10). Thus, the fiscal policy – here in the form of a given real transfer $-\tau_0^{-1}$ – determines the price level sequence $\{p_t\}_{t \geq 0}$.

10.7. Deficit finance with unbacked debt

Besides the transfer to the initial old agents $-\tau_0^{-1} \geq 0$, we now assume that the government purchases goods $g_t \geq 0$ in period $t \geq 0$. The government budget constraints are

$$\frac{B_1}{R_0} = g_0 - \tau_0^{-1}, \quad (10.7.1a)$$

$$\frac{B_{t+1}}{R_t} = g_t + B_t, \quad \forall t \geq 1. \quad (10.7.1b)$$

As in computation (10.6.2), we solve equations (10.7.1) forward to some date $T \geq 1$:

$$g_0 - \tau_0^{-1} = \frac{B_1}{R_0} = - \sum_{i=1}^T g_i \frac{1}{\prod_{j=0}^{i-1} R_j} + \frac{1}{\prod_{j=0}^{T-1} R_j} \frac{B_{T+1}}{R_T} \quad (10.7.2)$$

and, hence, government indebtedness in period $T \geq 1$ equals

$$\frac{B_{T+1}}{R_T} = g_T + \sum_{i=1}^{T-1} g_i \prod_{j=i}^{T-1} R_j + (g_0 - \tau_0^{-1}) \prod_{j=0}^{T-1} R_j, \quad (10.7.3)$$

which arises from the initial borrowing at time 0, $B_1/R_0 = g_0 - \tau_0^{-1}$, subsequent government expenditures $\{g_t\}_{t \geq 1}^T$, and compounded interest.

The addition of government purchases does not change the formulation of agents' optimization problems. Since government purchases do not enter agents' utility functions, agents' objective functions are unchanged. Budget sets are also unaffected: the budget constraint of an initial old agent is given by (10.6.6) and, at each date $t \geq 0$, a young agent faces budget constraints (10.6.4). Thus, an initial old agent continues to choose $c_0^{-1} = y_0^{-1} - \tau_0^{-1}$ and optimal decisions by an agent born at time $t \geq 0$ are summarized with the savings function in (10.5.7):

$$y_t^t - c_t^t = s(\alpha_t; y_t^t, y_{t+1}^t),$$

where $\alpha_t = 1/R_t$. We are led to the following:

DEFINITION: Given a sequence $\{g_t\}_{t=0}^\infty$ and a transfer $-\tau_0^{-1} \geq 0$, an equilibrium with bond-financed government deficits is a sequence $\{B_{t+1}, R_t\}_{t=0}^\infty$ such that (i) given R_t ,

$$B_{t+1} = \arg \max_b [u(y_t^t - b/R_t) + u(y_{t+1}^t + b)], \quad \forall t \geq 0; \quad (10.7.4)$$

and (ii) the government budget constraints (10.7.1) are satisfied.

Consumption of agents born in period $t \geq 0$ can be inferred from the sequence $\{B_{t+1}, R_t\}_{t=0}^{\infty}$ and consumption of the initial old agent is augmented by the transfer $-\tau_0^{-1}$. The variable B_{t+1} in the definition denotes both the solution to a representative agent's savings problem (10.7.4) and the quantity of bonds in government budget constraints (10.7.1), which implies that the bond market clears. Then by Walras' law, the goods market also clears:

$$c_t^t + c_t^{t-1} + g_t = y_t^t + y_t^{t-1}, \quad \forall t \geq 0. \quad (10.7.5)$$

To illustrate the equilibrium dynamics of the sequence $\{B_{t+1}, R_t\}_{t=0}^{\infty}$, we assume stationary endowments and government expenditures: $y_t^t = w_1$, $y_t^{t-1} = w_2 < w_1$, and $g_t = g$ for $t \geq 0$. For notational simplicity we define a savings function $f(R_t) = s(1/R_t; w_1, w_2)$. In an equilibrium, savings of the young $f(R_t)$ equals the value of one-period government bonds issued B_{t+1}/R_t . Substituting that equilibrium condition $B_{t+1} = R_t f(R_t)$ into government budget constraints (10.7.1), we obtain the following equations that fully characterize an equilibrium with bond-financed government deficits:

$$f(R_0) = g - \tau_0^{-1}, \quad (10.7.6a)$$

$$f(R_t) = g + R_{t-1}f(R_{t-1}), \quad \forall t \geq 1. \quad (10.7.6b)$$

Next, in the spirit of our earlier study of time 0 trading equilibria in section 10.3, we shall analyze stationary equilibria and then use the offer-curve approach of David Gale (1973) to compute nonstationary equilibria.

10.7.1. Stationary equilibria and the Laffer curve

Suppose that the economy has converged to a steady state. If so, equation (10.7.6b) would have a stationary solution, which is reasonable given that government purchases are constant and that $f(R_t)$ is time invariant (because endowments are time-invariant). Under the supposition that $R_t = R_{t-1} = R$, equation (10.7.6b) becomes

$$(1 - R)f(R) = g. \quad (10.7.7)$$

This equation lends credence to our earlier analogy that declining indebtedness in an unbacked-debt equilibrium can be likened to a government ‘amortizing’ its debt by the amount $(1 - R_{t-1})B_t/R_{t-1}$ in expression (10.6.7). We hinted that the government could instead use these ‘amortizations’ to make purchases of goods. In our present stationary equilibrium characterized by equation (10.7.7), indebtedness stays constant and, hence, the earlier ‘amortization’ on the left side is now fully used to purchase goods g on the right side. For later use, equation (10.7.7) yields a constant indebtedness equal to

$$\frac{B}{R} = f(R) = \frac{g}{1 - R}. \quad (10.7.8)$$

For example, suppose that $u(c) = \ln(c)$. Then $f(R) = \frac{w_1}{2} - \frac{w_2}{2R}$. We have graphed $(1 - R)f(R)$ in Figure 10.7.1. Notice that whenever a solution of equation (10.7.7) exists, there is at least one other (except at the very peak of the curve $(1 - R)f(R)$). When setting $g = 0$ in equation (10.7.7), we recover the pair of stationary equilibria that we have studied before. Specifically, in order for $(1 - R)f(R) = 0$, it must be true that either $R = 1$ or $f(R) = 0$. In the latter case of $f(R) = 0$, we are evaluating the savings function at an interest $R = R_L$ at which agents choose to consume their own endowments, i.e., our earlier low-interest-rate stationary equilibrium with a relative price of a good next period equal to $\alpha_L = 1/R_L$. In the other case of $R = 1$, it is our earlier high-interest-rate stationary equilibrium that we found to be Pareto optimal and could be supported by either fiat money or unbacked government debt. As evident from equation (10.7.7), this socially optimal interest rate $R = 1$ can no longer be supported in an equilibrium with bond-financed government deficits.

Given government purchases $g > 0$ in Figure 10.7.1, the interest rates in the two possible stationary equilibria are $R = \underline{R}$ and $R = \bar{R}$. While both stationary

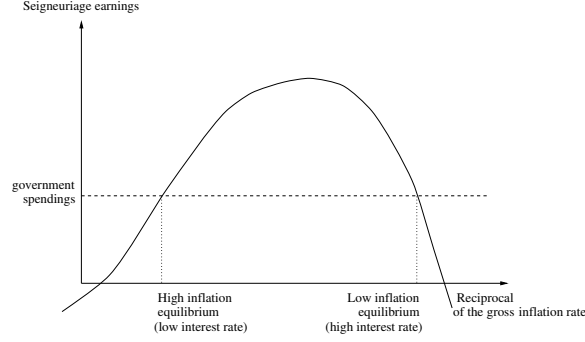


Figure 10.7.1: The Laffer curve in revenues from the inflation tax.

equilibria divert the same amount of goods g from private consumption, the equilibrium with the higher interest rate $\bar{R}$ imposes a smaller distortion on the intertemporal price. As we will elaborate when we study the isomorphic fiat-money equilibrium, $(1 - R)$ can be interpreted as a ‘tax rate’ and $f(R)$ as a ‘tax base’; thus, $(1 - R)f(R)$ becomes a Laffer curve showing ‘tax revenues’ in different stationary equilibria. Here the ‘tax rate’ increases when moving from right to left on the horizontal axis in Figure 10.7.1, so $\bar{R}$ is on the good side of the Laffer curve and $\underline{R}$ is on the bad side.

Keep in mind that Figure 10.7.1 depicts stationary equilibria and says nothing about how they were reached. So let’s turn to the analysis of nonstationary equilibria.

10.7.2. Nonstationary equilibria

To analyze nonstationary equilibria, we adopt the offer-curve approach of subsection 10.3.4. But instead of a clearinghouse, we retain the services of a Walrasian auctioneer who seeks an equilibrium price vector that ensures that the market clearing conditions (10.7.5) are satisfied. In the present context, the feasibility line becomes¹²

$$c_{t+1}^{t+1} + c_{t+1}^t = w_1 + w_2 - g, \quad \forall t \geq -1, \quad (10.7.9)$$

¹² Besides subtracting g from the right side of expression (10.7.9), we have added an extra initial feasibility line as compared to those in expression (10.3.9b) in subsection 10.3.4; namely,

as depicted in Figure 10.7.2. Notice that the endowment point (y_t^t, y_{t+1}^t) lies *outside* the feasibility line. The feasibility line is the only object that changes as compared to earlier Figure 10.3.1 in subsection 10.3.4. In particular, the offer curve remains the same for agents $t \geq 0$ and solves equations (10.3.7) for different values of the relative price $\alpha_t = 1/R_t$. The two intersections of the offer curve and the new feasibility line in Figure 10.7.2 constitute the two stationary equilibria with interest rate $\bar{R}$ and $\underline{R}$, respectively, as characterized above.

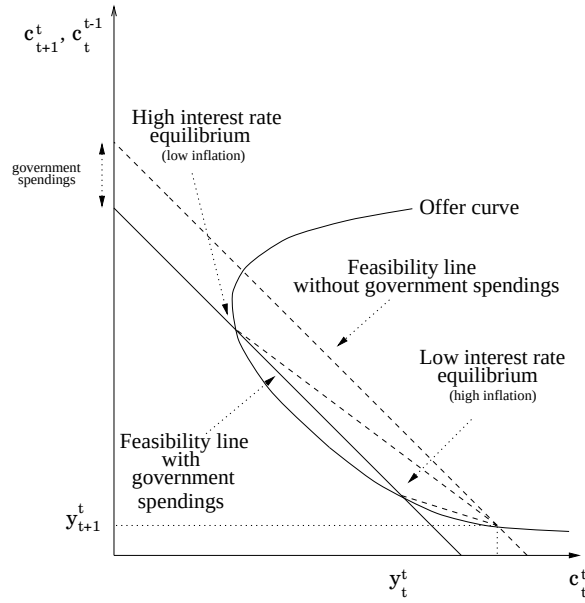


Figure 10.7.2: Equilibria with debt- or money-financed government deficit finance.

The steps laid out to solve for an equilibrium allocation in subsection 10.3.4 remain the same except for the very first period $t = 0$. Instead of treating c_0^0 as an initial condition, we deduce c_0^0 from the government's fiscal decisions. Specifically, as conveyed by equilibrium expression (10.7.6a), the government borrows to finance purchases of goods $g > 0$ and a transfer to the initial old

we now require that the consumption of the initial old and young agents in period 0 satisfy feasibility with equality, i.e., the goods market clears in period 0.

$-\tau_0^{-1} \geq 0$ on the right side, which equals the demand for savings by the young $f(R_0)$ on the left side. Thus, $c_0^0 = w_1 - g + \tau_0^{-1}$ ($= w_1 - B_1/R_0$) while the consumption of the initial old is $c_0^{-1} = w_2 - \tau_0^{-1}$.

As an illustration, suppose that $\tau_0^{-1} = 0$ and, hence, $c_0^0 = w_1 - g$. Given that initialization of c_0^0 , we can conduct a graphical analysis in Figure 10.7.2 similar to what we did in Figure 10.3.2 in subsection 10.3.5. We will then discover that the consumption allocation converges to the stationary equilibrium with interest rate $\underline{R}$. Convergence will still occur as we successively raise the initial transfer $-\tau_0^{-1}$ until it reaches the following critical value at which government indebtedness in period 0 becomes equal to that of the stationary equilibrium with the high interest rate $\bar{R}$:

$$-\tau_0^{-1} = f(\bar{R}) - g = \frac{g}{1 - \bar{R}} - g = \frac{\bar{R}g}{1 - \bar{R}}, \quad (10.7.10)$$

where we have invoked expression (10.7.8) for the level of government indebtedness in a stationary equilibrium. At the unique value of the transfer in expression (10.7.10), the economy starts and then remains in the stationary equilibrium with the high interest rate $\bar{R}$.

In summary, there is a continuum of solutions to difference equation (10.7.6), as indexed by the initial transfer $-\tau_0^{-1} \in [0, \bar{R}g/(1 - \bar{R})]$, all but one of which have $R_t \rightarrow \underline{R}$ as $t \rightarrow \infty$. Thus, the “bad” Laffer-curve equilibrium is stable.

Because net interest rates are negative, government indebtedness in any period $T \geq 1$ remains bounded, as shown in expression (10.7.3), where ‘weights’ on past expenditures decline geometrically with the time that has passed. This also means that the ‘weight’ on initial borrowing in period 0, $g_0 - \tau_0^{-1}$, will converge to zero. However, as we have learned, the initial transfer $-\tau_0^{-1}$ has a permanent impact when it determines whether the economy converges to a stationary equilibrium with a low or a high interest rate.

Solving the government budget constraint forward in expression (10.7.2) yields an unbounded present value of future expenditures, since discounting with negative net interest rates magnifies rather than dampens present values of future expenditures. The present value budget constraint nevertheless holds because bounded government indebtedness at any future date, as just described, becomes unbounded in present value. The negative present value of the stream of future government expenditures and the positive present value of outstanding unbacked government debt in the distant future both go to infinity at rates that allow the budget constraint to be satisfied.

10.8. Deficit finance with fiat money

Here we explore deficit finance with fiat money instead of the unbacked debt studied in the preceding section. The government budget constraints are then

$$\frac{M_0}{p_0} = g_0 - \tau_0^{-1}, \quad (10.8.1a)$$

$$\frac{M_t - M_{t-1}}{p_t} = g_t, \quad \forall t \geq 1. \quad (10.8.1b)$$

The equilibrium condition in the money market is $f(R_t) = M_t/p_t$, which can be written as $f(R_t) = M_{t-1}/p_t + (M_t - M_{t-1})/p_t$. The left side is the saving of the young. The term M_{t-1}/p_t on the right side is the dissaving of the old (the real value of currency that they exchange for time t consumption). The term $(M_t - M_{t-1})/p_t$ on the right side is the dissaving of the government (its deficit), which is the real value of the additional currency that the government prints at t and uses to purchase time t goods from the young.

DEFINITION: Given a sequence $\{g_t\}_{t=0}^\infty$ and a transfer $-\tau_0^{-1} \geq 0$, an equilibrium with money-financed government deficits is a sequence $\{M_t, p_t\}_{t=0}^\infty$ with $0 < p_t < +\infty$ and $M_t > 0$ such that (i) given $\{p_t, p_{t+1}\}$,

$$M_t = \arg \max_{m \geq 0} [u(y_t^t - m/p_t) + u(y_{t+1}^t + m/p_{t+1})], \quad \forall t \geq 0; \quad (10.8.2)$$

and (ii) the government budget constraints (10.8.1) are satisfied.

Notice that money neutrality prevails in this economy. That is, if there exists an equilibrium for some sequence $\{M_t, p_t\}_{t=0}^\infty$, then there are alternative sequences $\{\kappa M_t, \kappa p_t\}_{t=0}^\infty$, for any $\kappa \in (0, \infty)$, which also constitute equilibria with the same consumption allocation. This follows from the fact that the nominal money supply and the price level enter only as ratios of one another in all relevant equations.

10.8.1. Isomorphic equilibria under deficit finance

As we might anticipate from subsection 10.6.1, there are isomorphic equilibria of deficit finance where the government uses either fiat money or unbacked debt. That is, the government finances deficits either by printing money or by borrowing at a negative real net interest rate.

To go from an equilibrium with unbacked debt to an equilibrium with fiat money, pick some arbitrary nominal quantity of money M_0 and solve for p_0 from $M_0/p_0 = B_1/R_0$ ($= g_0 - \tau_0^{-1}$). All other prices, $\{p_t\}_{t \geq 1}$, are then recursively set according to $p_{t-1}/p_t = R_{t-1}$ for $t \geq 1$. Thereafter, nominal money supplies are set to satisfy $M_t/p_t = B_{t+1}/R_t$.

While our mapping into a fiat-money equilibrium imposes the same rates of return and amounts of assets as in the original unbacked-debt equilibrium, we still have to establish that government budget constraints (10.8.1) are satisfied. This is trivially true for constraint (10.8.1a) in period 0, given our choice of p_0 above. Regarding the remaining constraints (10.8.1b), we substitute our mapping into the left side of those budget constraints (to confirm that doing so yields the right side):

$$\frac{M_t - M_{t-1}}{p_t} = \frac{M_t}{p_t} - \frac{M_{t-1}}{p_t} \frac{p_{t-1}}{p_{t-1}} = \frac{B_{t+1}}{R_t} - \frac{B_t}{R_{t-1}} R_{t-1} = g_t$$

where the last equality invokes budget constraints (10.7.1b) in the original unbacked-debt equilibrium. Thus, we conclude that an equilibrium with bond-financed government deficits can be mapped into an equilibrium with fiat money replacing the unbacked debt.

We invite the reader to carry out the mapping in the opposite direction from an equilibrium with money-financed government deficits to one with unbacked government debt.

10.8.2. Computing equilibria again

To compute an equilibrium with a constant stream of government expenditures g , write equation (10.8.1b) as

$$\frac{M_t}{p_t} = \frac{M_{t-1}}{p_{t-1}} \frac{p_{t-1}}{p_t} + g, \quad \forall t \geq 1,$$

where M_0/p_0 is given by equation (10.8.1a). By substituting the equilibrium condition $M_t/p_t = f(R_t)$, where $R_t = p_t/p_{t+1}$, into these equations, we arrive at the same equilibrium characterization as that of an economy with bond-financed deficits in equations (10.7.6). Thus, the stationary-equilibrium characterization (10.7.7) remains the same:

$$(1 - R) f(R) = g \implies \frac{(p_{t+1}/p_t) - 1}{p_{t+1}/p_t} f\left(\frac{p_t}{p_{t+1}}\right) = g, \quad (10.8.3)$$

where we have used $R_t = p_t/p_{t+1}$ to illuminate the taxation analogy introduced in subsection 10.7.1. Recall that $(1 - R)$ was interpreted as a ‘tax rate’—here approximately equal to the net inflation rate $(p_{t+1}/p_t) - 1$ at low inflation—and $f(R)$ as a ‘tax base,’ namely agents’ demand for real money balances, which equals the real money supply M_t/p_t in equilibrium. Printing money causes inflation and thereby erodes the real value of money holdings; in a stationary equilibrium agents replenish nominal money holdings so that real balances remain unchanged. The government can thus be thought of as levying a ‘tax’ at a rate approximately equal to the net inflation rate on agents’ real money holdings, the ‘tax base.’ The ‘tax rate’ cannot exceed 100% (a money holder cannot fare worse than money becoming worthless), so the exact expression in equation (10.8.3) is $(1 - R) = [(p_{t+1}/p_t) - 1] / (p_{t+1}/p_t) < 1$.

Next, we revisit the example of logarithmic preferences from subsection 10.5.1. Recall difference equation (10.5.9), which we now reformulate using a lag operator,

$$\left(1 - \frac{w_2}{w_1} L^{-1}\right) p_t = \frac{2}{w_1} M_t, \quad (10.8.4)$$

where we now allow for a time-varying forcing term M_t . The solution to this difference equation is

$$p_t = \frac{2}{w_1} \sum_{i=0}^{\infty} \left(\frac{w_2}{w_1}\right)^i M_{t+i} + \kappa \left(\frac{w_1}{w_2}\right)^t. \quad (10.8.5)$$

If the nominal money supply stays constant, the solution simplifies to the earlier expression (10.5.10); otherwise the entire infinite sequence of nominal money supplies is required.

To illustrate such an equilibrium calculation, we proceed under the assumption that the economy has converged to a stationary equilibrium. Given the constant inflation rate $p_{t+1}/p_t = 1/R$ that characterizes a stationary equilibrium, a natural conjecture is that the nominal money supply grows at a constant rate, $M_{t+1} = zM_t$ for some $z > 0$. According to the first proposition in section 10.11, a necessary and sufficient condition for the existence of a monetary equilibrium is then $z u'(y_i^i)/u'(y_{i+1}^i) < 1$, which here translates to $z w_2/w_1 < 1$.¹³ Given the sequence $\{M_{t+i}\}_{i=0}^\infty = \{z^i M_t\}_{i=0}^\infty$ with $z < w_1/w_2$, we solve equation (10.8.5) to obtain price sequences consistent with money-market clearing,

$$p_t = \frac{2}{w_1} \frac{M_t}{1 - \frac{w_2}{w_1} z} + \kappa \left(\frac{w_1}{w_2} \right)^t. \quad (10.8.6)$$

Setting $\kappa = 0$ yields the price sequence that clears the money market in a stationary equilibrium in which the nominal money supply grows at a constant rate $z > 0$. The associated inflation rate is

$$\frac{p_{t+1}}{p_t} = \frac{\frac{2}{w_1} \frac{M_{t+1}}{1 - \frac{w_2}{w_1} z}}{\frac{2}{w_1} \frac{M_t}{1 - \frac{w_2}{w_1} z}} = \frac{M_{t+1}}{M_t} = z, \quad (10.8.7)$$

and the real money supply is

$$\frac{M_t}{p_t} = \frac{M_t}{\frac{2}{w_1} \frac{M_t}{1 - \frac{w_2}{w_1} z}} = \frac{w_1 - w_2 z}{2} = f(1/z), \quad (10.8.8)$$

where the last equality is the savings function evaluated at the rate of return $R = p_t/p_{t+1} = 1/z$.

¹³ As detailed in section 10.11, Wallace (1980) shows sufficiency of the stated condition by proving the existence of a stationary monetary equilibrium, which is a counterpart to the stationary solution of our difference equation (10.8.6). While Wallace replaces our unfunded government expenditures g with lump-sum transfers to agents, that difference in assumptions does not affect Wallace's proofs of necessity and sufficiency for the existence of at least one monetary equilibrium. (The difference in assumptions does, of course, matter for any welfare analysis.)

In a stationary equilibrium with money-financed government deficits equal to g , the growth rate of the nominal money supply z must satisfy equilibrium condition (10.8.3):

$$\frac{(p_{t+1}/p_t) - 1}{p_{t+1}/p_t} f\left(\frac{p_t}{p_{t+1}}\right) = g \implies \frac{z-1}{z} f\left(\frac{1}{z}\right) = g. \quad (10.8.9)$$

As depicted in Figure 10.7.1, there are two values of z satisfying equation (10.8.9): a high-inflation-rate equilibrium with $z = 1/\underline{R} \gg 1$ and a low-inflation-rate equilibrium with $z = 1/\bar{R} > 1$. Drawing on the isomorphism established in subsection 10.8.1—that equilibria are identical whether the government finances its deficits by printing money or by borrowing at a negative real net interest rate—we can apply the analysis of bond-financed equilibria in subsection 10.7.2. The low-inflation-rate stationary equilibrium is attained in period 0 by setting the initial transfer $-\tau_0^{-1}$ equal to the critical value in equation (10.7.10). If the initial transfer is smaller, however, all monetary equilibria converge to the high-inflation-rate stationary equilibrium. As noted earlier, the “bad” Laffer-curve equilibrium is stable.

The stability of the bad Laffer-curve equilibrium is a feature of perfect foresight dynamics. It is natural to ask whether better outcomes can emerge from an alternative to perfect foresight dynamics. Bruno and Fischer (1990) and Marcet and Sargent (1989) analyze how the system behaves under two different types of imperfect foresight, adaptive dynamics. They find that under either a crude form of adaptive expectations or a least-squares learning scheme, R_t converges to $\bar{R}$. This finding is reassuring because the comparative dynamics are more plausible at $\bar{R}$ (larger deficits bring higher inflation). Furthermore, Marimon and Sunder (1993) present experimental evidence supporting the equilibrium selection implied by adaptive dynamics. Marcet and Nicolini (2003) build and calibrate an adaptive model of several Latin American hyperinflations based on this equilibrium selection. Sargent, Williams, and Zha (2009) extend and estimate that model.

10.9. Within-generation heterogeneity

This section describes an overlapping generations model having within-generation heterogeneity of endowments. We assume that within each generation born at $t \geq 0$, there are J groups of agents. There is a constant number N_j of group j agents. Agents of group j are endowed with $w_1(j)$ when young and $w_2(j)$ when old. For a consumer group j born at time t , each young person's behavior is summarized by

$$s_{jt} = f(R_t, j) = \arg \max_s [u(w_1(j) - s) + u(w_2(j) + R_t s)]. \quad (10.9.1)$$

If we assume that all consumers of generation t have preferences $U^t(\mathbf{c}^t) = \ln(c_t^t) + \ln(c_{t+1}^t)$, the saving function is

$$f(R_t, j) = .5 \left(w_1(j) - \frac{w_2(j)}{R_t} \right).$$

At $t = 0$, there are old people who are endowed in the aggregate with H units of an inconvertible currency.

For example, assume that $J = 2$, that $(w_1(1), w_2(1)) = (\alpha, 0)$ and that $(w_1(2), w_2(2)) = (0, \beta)$, where $\alpha > 0$, $\beta > 0$. The type 1 people are lenders, while the type 2 are borrowers. For the case of log preference we have the savings functions $f(R, 1) = \alpha/2$, $f(R, 2) = -\beta/(2R)$.

10.9.1. Nonmonetary equilibrium

A nonmonetary equilibrium consists of sequences of rates of return, $\{R_t\}_{t=0}^\infty$, and savings rates, $\{s_{jt}\}_{t=0}^\infty$ for $j = 1, \dots, J$, that satisfy (1) $s_{jt} = f(R_t, j)$, and (2) $\sum_{j=1}^J N_j f(R_t, j) = 0$. Condition (1) builds in consumer optimization; condition (2) says that aggregate net savings equals zero (borrowing equals lending).

For the case in which the endowments, preferences, and group sizes are constant across time, the interest rate is determined at the intersection of the aggregate savings function with the R axis, depicted as R_1 in Figure 10.9.1. No intergenerational transfers occur in the nonmonetary equilibrium. The equilibrium consists of a sequence of separate two-period pure consumption loan economies of a type analyzed by Irving Fisher (1907).

10.9.2. Monetary equilibrium

In an equilibrium with valued fiat currency, at each date $t \geq 0$ the old receive goods from the young in exchange for the currency stock H .¹⁴ An equilibrium with valued fiat money is a set of sequences of price levels, $\{p_t\}_{t=0}^{\infty}$, rates of return, $\{R_t\}_{t=0}^{\infty}$, and savings rates, $\{s_{jt}\}_{t=0}^{\infty}$ for $j = 1, \dots, J$, such that (1) $p_t > 0$, (2) $R_t = p_t/p_{t+1}$, (3) $s_{jt} = f(R_t, j)$, and (4) $\sum_{j=1}^J N_j f(R_t, j) = \frac{H}{p_t}$. Condition (1) states that currency is valued at all dates. Condition (2) states that currency and consumption loans are perfect substitutes. Condition (3) requires that saving decisions are optimal. Condition (4) equates the net saving of the young (the left side) to the net dissaving of the old (the right side). The old supply currency inelastically.

We can determine a stationary equilibrium graphically. A stationary equilibrium satisfies $p_t = p$ for all t , which implies $R = 1$ for all t . Thus, if it exists, a stationary equilibrium solves

$$\sum_{j=1}^J N_j f(1, j) = \frac{H}{p} \quad (10.9.2)$$

for a positive price level. (See Figure 10.9.1.) Evidently, a stationary monetary equilibrium exists if the net savings of the young are positive at $R = 1$.

For the special case of logarithmic preferences and two classes of young people, the aggregate savings function of the young is time invariant and equal to

$$\sum_j f(R, j) = .5(N_1\alpha - N_2\frac{\beta}{R}).$$

Note that the equilibrium condition (10.9.2) can be written

$$.5N_1\alpha = .5N_2\frac{\beta}{R} + \frac{H}{p}.$$

The left side is the demand for savings or the demand for “currency” while the right side is the supply, consisting of privately issued IOU’s (the first term) and government-issued currency. The right side is thus an abstract version of what is called M1, which is a sum of privately issued IOUs (demand deposits) and government-issued reserves and currency.

¹⁴ As before, we assume that the initial old agents start out owning the currency stock H . The distribution of currency across the initial old agents is immaterial to our analysis here.

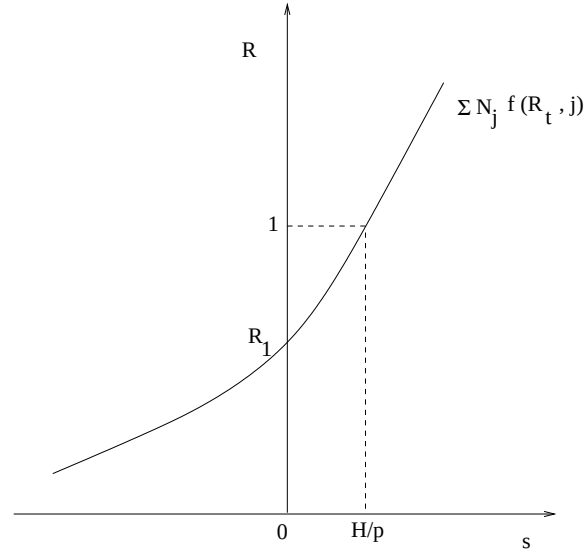


Figure 10.9.1: The intersection of the aggregate savings function with a horizontal line at $R = 1$ determines a stationary equilibrium value of the price level, if positive.

10.9.3. Nonstationary equilibria

Mathematically, the equilibrium conditions for the model with log preferences and two groups have the same structure as the model analyzed previously in equations (10.5.9) and (10.5.10), with simple reinterpretations of parameters. We leave it to the reader here and in an exercise to show that if there exists a stationary equilibrium with valued fiat currency, then there exists a continuum of equilibria with valued fiat currency, all but one of which have the real value of government currency approaching zero asymptotically. A linear difference equation like (10.5.9) supports this conclusion.

10.10. Additional topics

10.10.1. Lucas tree

To set up an environment with a “Lucas tree,” take the preference and endowment structure to be the same as subsection 10.3.2 and modify only one feature. Endow the initial old with a claim to a constant stream of $d > 0$ units of consumption for each $t \geq 0$.¹⁵ Thus, the budget constraint of an initial old person now becomes

$$q_0^0 c_0^{-1} = d \sum_{t=0}^{\infty} q_t^0 + q_0^0 y_0^{-1} = d \sum_{t=0}^{\infty} q_t^0 + q_0^0 \epsilon.$$

The offer curve of each young agent remains as before, but now the feasibility line is

$$c_i^i + c_i^{i-1} = y_i^i + y_i^{i-1} + d = 1 + d$$

for all $i \geq 0$. Note that young agents are endowed below the feasibility line in Figure 10.10.1. It seems that there are two candidates for stationary equilibria, one with constant $\alpha < 1$, another with constant $\alpha > 1$. The one with $\alpha < 1$ is associated with the steeper budget line in Figure 10.10.1. However, the candidate stationary equilibrium with $\alpha > 1$ cannot be an equilibrium. At the price system associated with an $\alpha > 1$, the wealth of the initial old would be unbounded, which would prompt them to consume an unbounded amount, which is not feasible. This argument rules out not only the stationary $\alpha > 1$ equilibrium but also all nonstationary candidate equilibria that converge to that constant α . Therefore, there is a unique equilibrium; it is stationary and has $\alpha < 1$.

If we interpret the gross rate of return on the tree as $\alpha^{-1} = \frac{p+d}{p}$, where $p = \sum_{t=1}^{\infty} q_t^0 d$, we can compute that $p = \frac{d}{R-1}$ where $R = \alpha^{-1}$. Here p is the ex-dividend price of the Lucas tree for all $t \geq 0$ in a sequential-trading equilibrium.

For the sake of argument, suppose that the Lucas tree bears no fruit, $d = 0$, but the tree itself is durable. What would be the price of such a tree? According

¹⁵ This is a version of an example of Brock (1990). The ‘Lucas tree’ refers to a colorful interpretation of a dividend stream as ‘fruit’ falling from a ‘tree’ in a pure exchange economy studied by Lucas (1978). See chapter 14.

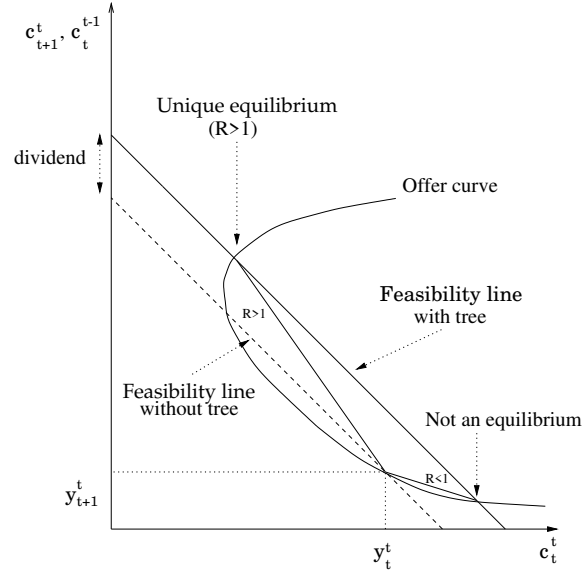


Figure 10.10.1: Unique equilibrium with a fixed-dividend asset.

to the above price formula based on ‘fundamentals’, $p = \sum_{t=1}^{\infty} q_t^0 d$, the tree would be worthless, $p = 0$. However, if the tree takes on the same role as the constant quantity of fiat money in section 10.5, the price of the tree would be governed by the very same calculations of valuation. Though, please note that the price level in section 10.5 is the price of goods in terms of fiat money, which in our current context would be the price of goods in terms of Lucas tree. Hence, the price of Lucas tree (in terms of goods) would be the inverse of the price level in section 10.5.

10.10.2. Population growth

It is easy to extend these models to the case in which there is growth in the population. Let there be $N_t = nN_{t-1}$ identical young people at time t , with $n > 0$. For example, the extended budget constraint (10.4.1) in period $t \geq 0$ becomes

$$\frac{N_t m_t^t - N_{t-1} m_{t-1}^{t-1}}{p_t} + N_{t-1} \tau_t^{t-1} + N_t \tau_t^t + \frac{N_t b_{t+1}^t}{R_t} = N_t g_t + N_{t-1} b_t^{t-1},$$

where g_t is per-young-person government purchases. Dividing both sides of the budget constraint by N_t and rearranging gives

$$\hat{m}_t + n^{-1} \tau_t^{t-1} + \tau_t^t + \frac{b_{t+1}^t}{R_t} = n^{-1} \hat{m}_{t-1} \frac{p_{t-1}}{p_t} + g_t + n^{-1} b_t^{t-1}, \quad (10.10.1)$$

where $\hat{m}_i$ is real money balances of a young agent at time i , $\hat{m}_i \equiv m_i^i / p_i$.

As a comparison, recall the high-interest-rate stationary equilibrium with unbacked debt in section 10.6 with interest rate $R = 1$. The counterpart in the present economy with population growth is a stationary equilibrium in which the amount of bonds purchased by a young agent stays constant over time, $b_{i+1}^i = b$ for all i , and satisfies the corresponding version of government budget constraint (10.10.1),

$$\frac{b}{R_t} = n^{-1} b.$$

Evidently, the stationary high interest rate is now $R = n$. Similarly, the counterpart to the stationary equilibrium with fiat money in section 10.5 has now constant per-young-agent real money holdings, $\hat{m}_i = \hat{m}$ for all i , that satisfy the corresponding version of government budget constraint (10.10.1),

$$\hat{m} = n^{-1} \hat{m} \frac{p_{t-1}}{p_t}.$$

Thus, the stationary return on fiat money, $p_{t-1}/p_t = n$, is the same as the return on unbacked debt, $R = n$, which was to be expected since the two equilibria can be shown to be isomorphic (as we showed for $n = 1$ in subsection 10.6.1).

10.10.3. Gift-giving equilibrium

Michihiro Kandori (1992) and Lones Smith (1992) have used ideas from the literature on reputation (see chapter 25) to study whether there exist history-dependent sequences of gifts that support an optimal allocation. Their idea is to set up the economy as a game played with a sequence of players. We briefly describe a gift-giving game for an overlapping generations economy in which voluntary intergenerational gifts support an optimal allocation. Suppose that the consumption of an initial old person is

$$c_0^{-1} = y_0^{-1} + s_0$$

and the utility of each young agent is

$$u(y_i^i - s_i) + u(y_{i+1}^i + s_{i+1}), \quad i \geq 0 \quad (10.10.2)$$

where $s_i \geq 0$ is the gift from a young person at i to an old person at i . Suppose that the endowment pattern is $y_i^i = 1 - \epsilon$, $y_{i+1}^i = \epsilon$, where $\epsilon \in (0, .5)$.

Consider the following system of expectations, to which a young person chooses whether to conform:

$$s_i = \begin{cases} .5 - \epsilon & \text{if } v_i = \bar{v}; \\ 0 & \text{otherwise.} \end{cases} \quad (10.10.3a)$$

$$v_{i+1} = \begin{cases} \bar{v} & \text{if } v_i = \bar{v} \text{ and } s_i = .5 - \epsilon; \\ \underline{v} & \text{otherwise.} \end{cases} \quad (10.10.3b)$$

Here we are free to take $\bar{v} = 2u(.5)$ and $\underline{v} = u(1 - \epsilon) + u(\epsilon)$. These are “promised utilities.” We make them serve as “state variables” that summarize the history of intergenerational gift giving. To start, we need an initial value v_0 . Equations (10.10.3) act as the transition laws that young agents face in choosing s_i in (10.10.2).

An initial condition v_0 and the rule (10.10.3) form a system of expectations that tells the young person of each generation what he is expected to give. His gift is immediately handed over to an old person. A system of expectations is called an *equilibrium* if for each $i \geq 0$, each young agent chooses to conform.

We can immediately compute two equilibrium systems of expectations. The first is the “autarky” equilibrium: give nothing yourself and expect all future generations to give nothing. To represent this equilibrium within equations (10.10.3), set $v_0 \neq \bar{v}$. It is easy to verify that each young person will confirm

what is expected of him in this equilibrium. Given that future generations will not give, each young person chooses not to give.

For the second equilibrium, set $v_0 = \bar{v}$. Here each consumer chooses to give the expected amount, because failure to do so causes the next generation of young people not to give; whereas affirming the expectation to give passes that expectation along to the next generation, which affirms it in turn. Each of these equilibria is credible, in the sense of subgame perfection, to be studied extensively in chapter 25.

Narayana Kocherlakota (1998) has compared gift giving and monetary equilibria in a variety of environments and has used the comparison to provide a precise sense in which “money” substitutes for “memory”.

10.11. Optimality and existence of monetary equilibria

Wallace (1980) discusses a connection between nonoptimality of the equilibrium without valued money and the existence of monetary equilibria. For an economy not having the storage technology available in Wallace’s economy, we study how arguments similar to his apply. The environment is as follows. At any date t , the population consists of N_t young agents and N_{t-1} old agents where $N_t = nN_{t-1}$ with $n > 0$. Each young person is endowed with $y_1 > 0$ goods, and an old person receives the endowment $y_2 > 0$. Preferences of a young agent at time t are given by the utility function $u(c_t^t, c_{t+1}^t)$, which is twice differentiable with indifference curves that are convex to the origin. The two goods in the utility function are normal goods, and

$$\theta(c_1, c_2) \equiv u_1(c_1, c_2)/u_2(c_1, c_2),$$

the marginal rate of substitution function, approaches infinity as c_2/c_1 approaches infinity and approaches zero as c_2/c_1 approaches zero. The welfare of the initial old agents at time 0 is strictly increasing in c_0^{-1} , and each one of them is endowed with y_2 goods and $m_{-1}^{-1} > 0$ units of fiat money. Thus, the beginning-of-period aggregate nominal money balances in the initial period 0 are $M_{-1} = N_{-1}m_{-1}^{-1}$.

For all $t \geq 0$, M_t , the post-transfer time t stock of fiat money obeys $M_t = zM_{t-1}$ with $z > 0$. The time t transfer (or tax), $(z - 1)M_{t-1}$, is divided

equally at time t among the N_{t-1} members of the current old generation. The transfers (or taxes) are fully anticipated and are viewed as lump-sum: they do not depend on consumption and saving behavior. The budget constraints of a young agent born in period $t \geq 0$ are

$$c_t^t + \frac{m_t^t}{p_t} \leq y_1, \quad (10.11.1)$$

$$c_{t+1}^t \leq y_2 + \frac{m_t^t}{p_{t+1}} + \frac{(z-1)}{N_t} \frac{M_t}{p_{t+1}}, \quad (10.11.2)$$

$$m_t^t \geq 0, \quad (10.11.3)$$

where $p_t > 0$ is the time t price level. In a nonmonetary equilibrium, the price level is infinite, so the real values of both money holdings and transfers are zero. Since all members in a generation are identical, the nonmonetary equilibrium is autarky with a marginal rate of substitution equal to

$$\theta_{\text{aut}} \equiv \frac{u_1(y_1, y_2)}{u_2(y_1, y_2)}.$$

We ask two questions about this economy. Under what circumstances does a monetary equilibrium exist? And, when it exists, under what circumstances does it improve matters?

Let $\hat{m}_t$ denote the equilibrium real money balances of a young agent at time t , $\hat{m}_t \equiv M_t/(N_t p_t)$. Substitution of equilibrium money holdings into budget constraints (10.11.1) and (10.11.2) at equality yield $c_t^t = y_1 - \hat{m}_t$ and $c_{t+1}^t = y_2 + n\hat{m}_{t+1}$. In a monetary equilibrium, $\hat{m}_t > 0$ for all t and the marginal rate of substitution $\theta(c_t^t, c_{t+1}^t)$ satisfies

$$\theta(y_1 - \hat{m}_t, y_2 + n\hat{m}_{t+1}) = \frac{p_t}{p_{t+1}} > \theta_{\text{aut}}, \quad \forall t \geq 0. \quad (10.11.4)$$

The equality part of (10.11.4) is the first-order condition for money holdings of an agent born in period t evaluated at the equilibrium allocation. Since $c_t^t < y_1$ and $c_{t+1}^t > y_2$ in a monetary equilibrium, the inequality in (10.11.4) follows from the assumption that the two goods in the utility function are normal goods.

Another useful characterization of the equilibrium rate of return on money, p_t/p_{t+1} , can be obtained as follows. By the rule generating M_t and the equilibrium condition $M_t/p_t = N_t \hat{m}_t$, we have for all t ,

$$\frac{p_t}{p_{t+1}} = \frac{M_{t+1}}{z M_t} \frac{p_t}{p_{t+1}} = \frac{N_{t+1} \hat{m}_{t+1}}{z N_t \hat{m}_t} = \frac{n}{z} \frac{\hat{m}_{t+1}}{\hat{m}_t}. \quad (10.11.5)$$

We are now ready to address our first question, under what circumstances does a monetary equilibrium exist?

PROPOSITION: $\theta_{\text{aut}} z < n$ is necessary and sufficient for the existence of at least one monetary equilibrium.

PROOF: We first establish necessity. Suppose to the contrary that there is a monetary equilibrium and $\theta_{\text{aut}} z/n \geq 1$. Then by the inequality part of (10.11.4) and expression (10.11.5), we have for all t ,

$$\frac{\hat{m}_{t+1}}{\hat{m}_t} > \frac{z\theta_{\text{aut}}}{n} \geq 1. \quad (10.11.6)$$

If $z\theta_{\text{aut}}/n > 1$, one plus the net growth rate of $\hat{m}_t$ is bounded uniformly above one and, hence, the sequence $\{\hat{m}_t\}$ is unbounded, which is inconsistent with an equilibrium because real money balances per capita cannot exceed the endowment y_1 of a young agent. If $z\theta_{\text{aut}}/n = 1$, the strictly increasing sequence $\{\hat{m}_t\}$ in (10.11.6) might not be unbounded but converge to some constant $\hat{m}_\infty$. According to (10.11.4) and (10.11.5), the marginal rate of substitution will then converge to n/z , which by assumption is now equal to θ_{aut} , the marginal rate of substitution in autarky. Thus, real balances must be zero in the limit, which contradicts the existence of a strictly increasing sequence of positive real balances in (10.11.6).

To show sufficiency, we prove the existence of a unique equilibrium with constant per capita real money balances when $\theta_{\text{aut}} z < n$. Substitute our candidate equilibrium, $\hat{m}_t = \hat{m}_{t+1} \equiv \hat{m}$, into (10.11.4) and (10.11.5), which yields two equilibrium conditions,

$$\theta(y_1 - \hat{m}, y_2 + n\hat{m}) = \frac{n}{z} > \theta_{\text{aut}}.$$

The inequality part is satisfied under the parameter restriction of the proposition, and we only have to show the existence of $\hat{m} \in [0, y_1]$ that satisfies the equality part. But the existence (and uniqueness) of such a $\hat{m}$ is trivial. Note that the marginal rate of substitution on the left side of the equality is equal to θ_{aut} when $\hat{m} = 0$. Next, our assumptions on preferences imply that the marginal rate of substitution is strictly increasing in $\hat{m}$, and approaches infinity when $\hat{m}$ approaches y_1 . ■

The stationary monetary equilibrium in the proof will be referred to as the $\hat{m}$ equilibrium. In general, there are other nonstationary monetary equilibria

when the parameter condition of the proposition is satisfied. For example, in the case of logarithmic preferences and a constant population, recall the continuum of equilibria indexed by the scalar $\kappa > 0$ in expression (10.5.10). But here we choose to focus solely on the stationary $\hat{m}$ equilibrium and its welfare implications. The $\hat{m}$ equilibrium will be compared to other feasible allocations using the Pareto criterion. Evidently, an allocation $C = \{c_0^{-1}; (c_t^t, c_{t+1}^t), t \geq 0\}$ is feasible if

$$N_t c_t^t + N_{t-1} c_t^{t-1} \leq N_t y_1 + N_{t-1} y_2, \quad \forall t \geq 0,$$

or, equivalently,

$$n c_t^t + c_t^{t-1} \leq n y_1 + y_2, \quad \forall t \geq 0. \quad (10.11.7)$$

The definition of Pareto optimality is:

DEFINITION: A feasible allocation C is Pareto optimal if there is no other feasible allocation $\tilde{C}$ such that

$$\begin{aligned} \tilde{c}_0^{-1} &\geq c_0^{-1}, \\ u(\tilde{c}_t^t, \tilde{c}_{t+1}^t) &\geq u(c_t^t, c_{t+1}^t), \quad \forall t \geq 0, \end{aligned}$$

and at least one of these weak inequalities holds with strict inequality.

We first examine under what circumstances the nonmonetary equilibrium (autarky) is Pareto optimal.

PROPOSITION: $\theta_{\text{aut}} \geq n$ is necessary and sufficient for the optimality of the nonmonetary equilibrium (autarky).

PROOF: To establish sufficiency, suppose to the contrary that there exists another feasible allocation $\tilde{C}$ that is Pareto superior to autarky and $\theta_{\text{aut}} \geq n$. Without loss of generality, assume that the allocation $\tilde{C}$ satisfies (10.11.7) with equality. (Given an allocation that is Pareto superior to autarky but that does not satisfy (10.11.7), one can easily construct another allocation that is Pareto superior to the given allocation, and hence to autarky.) Let period t be the first period when this alternative allocation $\tilde{C}$ differs from the autarkic allocation. The requirement that the old generation in this period is not made worse off, $\tilde{c}_t^{t-1} \geq y_2$, implies that the first perturbation from the autarkic allocation must be $\tilde{c}_t^t < y_1$, with the subsequent implication that $\tilde{c}_{t+1}^t > y_2$. It follows that

the consumption of young agents at time $t + 1$ must also fall below y_1 , and we define

$$\epsilon_{t+1} \equiv y_1 - \tilde{c}_{t+1}^{t+1} > 0. \quad (10.11.8)$$

Now, given $\tilde{c}_{t+1}^{t+1}$, we compute the smallest number $\tilde{c}_{t+2}^{t+1}$ that satisfies

$$u(\tilde{c}_{t+1}^{t+1}, \tilde{c}_{t+2}^{t+1}) \geq u(y_1, y_2).$$

Let $\tilde{c}_{t+2}^{t+1}$ be the solution to this problem. Since the allocation $\tilde{C}$ is Pareto superior to autarky, we have $\tilde{c}_{t+2}^{t+1} \geq \bar{c}_{t+2}^{t+1}$. Before using this inequality, though, we want to derive a convenient expression for $\tilde{c}_{t+2}^{t+1}$.

Consider the indifference curve of $u(c_1, c_2)$ that yields a fixed utility equal to $u(y_1, y_2)$. In general, along an indifference curve, $c_2 = h(c_1)$, where $h' = -u_1/u_2 = -\theta$ and $h'' > 0$. Therefore, applying the intermediate value theorem to h , we have

$$h(c_1) = h(y_1) + (y_1 - c_1)[-h'(y_1) + f(y_1 - c_1)], \quad (10.11.9)$$

where the function f is strictly increasing and $f(0) = 0$.

Now, since $(\tilde{c}_{t+1}^{t+1}, \tilde{c}_{t+2}^{t+1})$ and (y_1, y_2) are on the same indifference curve, we can use (10.11.8) and (10.11.9) to write

$$\tilde{c}_{t+2}^{t+1} = y_2 + \epsilon_{t+1}[\theta_{\text{aut}} + f(\epsilon_{t+1})],$$

and after invoking $\tilde{c}_{t+2}^{t+1} \geq \bar{c}_{t+2}^{t+1}$, we have

$$\tilde{c}_{t+2}^{t+1} - y_2 \geq \epsilon_{t+1}[\theta_{\text{aut}} + f(\epsilon_{t+1})]. \quad (10.11.10)$$

Since $\tilde{C}$ satisfies (10.11.7) at equality, we also have

$$\epsilon_{t+2} \equiv y_1 - \tilde{c}_{t+2}^{t+2} = \frac{\tilde{c}_{t+2}^{t+1} - y_2}{n}. \quad (10.11.11)$$

Substitution of (10.11.10) into (10.11.11) yields

$$\begin{aligned} \epsilon_{t+2} &\geq \epsilon_{t+1} \frac{\theta_{\text{aut}} + f(\epsilon_{t+1})}{n} \\ &> \epsilon_{t+1}, \end{aligned} \quad (10.11.12)$$

where the strict inequality follows from $\theta_{\text{aut}} \geq n$ and $f(\epsilon_{t+1}) > 0$ (implied by $\epsilon_{t+1} > 0$). Continuing these computations of successive values of ϵ_{t+k} yields

$$\epsilon_{t+k} \geq \epsilon_{t+1} \prod_{j=1}^{k-1} \frac{\theta_{\text{aut}} + f(\epsilon_{t+j})}{n} > \epsilon_{t+1} \left[\frac{\theta_{\text{aut}} + f(\epsilon_{t+1})}{n} \right]^{k-1}, \text{ for } k > 2,$$

where the strict inequality follows from the fact that $\{\epsilon_{t+j}\}$ is a strictly increasing sequence. Thus, the ϵ sequence is bounded below by a strictly increasing exponential and hence is unbounded. But such an unbounded sequence violates feasibility because ϵ cannot exceed the endowment y_1 of a young agent. It follows that we can rule out the existence of a Pareto superior allocation $\tilde{C}$, and conclude that $\theta_{\text{aut}} \geq n$ is a sufficient condition for the optimality of autarky.

To establish necessity, we prove the existence of an alternative feasible allocation $\hat{C}$ that is Pareto superior to autarky when $\theta_{\text{aut}} < n$. First, pick an $\epsilon > 0$ sufficiently small so that

$$\theta_{\text{aut}} + f(\epsilon) \leq n, \quad (10.11.13)$$

where f is defined implicitly by equation (10.11.9). Second, set $\hat{c}_t^t = y_1 - \epsilon \equiv \hat{c}_1$, and

$$\hat{c}_{t+1}^t = y_2 + \epsilon[\theta_{\text{aut}} + f(\epsilon)] \equiv \hat{c}_2, \quad \forall t \geq 0. \quad (10.11.14)$$

That is, we have constructed a consumption bundle $(\hat{c}_1, \hat{c}_2)$ that lies on the same indifference curve as (y_1, y_2) , and from (10.11.13) and (10.11.14), we have

$$\hat{c}_2 \leq y_2 + n\epsilon,$$

which ensures that the condition for feasibility (10.11.7) is satisfied for $t \geq 1$. By setting $\hat{c}_0^{-1} = y_2 + n\epsilon$, feasibility is also satisfied in period 0 and the initial old generation is then strictly better off under the alternative allocation $\hat{C}$. ■

With a constant nominal money supply, $z = 1$, the two propositions show that a monetary equilibrium exists if and only if the nonmonetary equilibrium is suboptimal. In that case, the following proposition establishes that the stationary $\hat{m}$ equilibrium is optimal.

PROPOSITION: Given $\theta_{\text{aut}}z < n$, then $z \leq 1$ is necessary and sufficient for the optimality of the stationary monetary equilibrium $\hat{m}$.

PROOF: The class of feasible stationary allocations with $(c_t^t, c_{t+1}^t) = (c_1, c_2)$ for all $t \geq 0$, is given by

$$c_1 + \frac{c_2}{n} \leq y_1 + \frac{y_2}{n}, \quad (10.11.15)$$

i.e., the condition for feasibility in (10.11.7). It follows that the $\hat{m}$ equilibrium satisfies (10.11.15) at equality, and we denote the associated consumption allocation of an agent born at time $t \geq 0$ by $(\hat{c}_1, \hat{c}_2)$. It is also the case that

$(\hat{c}_1, \hat{c}_2)$ maximizes an agent's utility subject to budget constraints (10.11.1) and (10.11.2). The consolidation of these two constraints yields

$$c_1 + \frac{z}{n}c_2 \leq y_1 + \frac{z}{n}y_2 + \frac{z(z-1)}{n} \frac{M_t}{N_t} \frac{1}{p_{t+1}}, \quad (10.11.16)$$

where we have used the stationary rate of return in (10.11.5), $p_t/p_{t+1} = n/z$. After also invoking $zM_t = M_{t+1}$, $n = N_{t+1}/N_t$, and the equilibrium condition $M_{t+1}/(p_{t+1}N_{t+1}) = \hat{m}$, expression (10.11.16) simplifies to

$$c_1 + \frac{z}{n}c_2 \leq y_1 + \frac{z}{n}y_2 + (z-1)\hat{m}. \quad (10.11.17)$$

To prove the statement about necessity, Figure 10.11.1 depicts the two curves (10.11.15) and (10.11.17) when condition $z \leq 1$ fails to hold, i.e., we assume that $z > 1$. The point that maximizes utility subject to (10.11.15) is denoted $(\bar{c}_1, \bar{c}_2)$. Transitivity of preferences and the fact that the slope of budget line (10.11.17) is flatter than that of (10.11.15) imply that $(\hat{c}_1, \hat{c}_2)$ lies southeast of $(\bar{c}_1, \bar{c}_2)$. By revealed preference, then, $(\bar{c}_1, \bar{c}_2)$ is preferred to $(\hat{c}_1, \hat{c}_2)$ and all generations born in period $t \geq 0$ are better off under the allocation $\bar{C}$. The initial old generation can also be made better off under this alternative allocation since it is feasible to strictly increase their consumption,

$$\bar{c}_0^{-1} = y_2 + n(y_1 - \bar{c}_0^0) > y_2 + n(y_1 - \hat{c}_0^0) = \hat{c}_0^{-1}.$$

Thus, we have established that $z \leq 1$ is necessary for the optimality of the stationary monetary equilibrium $\hat{m}$.

To prove sufficiency, note that (10.11.4), (10.11.5) and $z \leq 1$ imply that

$$\theta(\hat{c}_1, \hat{c}_2) = \frac{n}{z} \geq n.$$

We can then construct an argument that is analogous to the sufficiency part of the proof to the preceding proposition. ■

As pointed out by Wallace (1980), the proposition implies no connection between the path of the price level in an $\hat{m}$ equilibrium and the optimality of that equilibrium. Thus, there may be an optimal monetary equilibrium with positive inflation, for example, if $\theta_{\text{aut}} < n < z \leq 1$; and there may be a nonoptimal monetary equilibrium with a constant price level, for example, if $z = n > 1 > \theta_{\text{aut}}$. What counts is the nominal quantity of fiat money. The proposition suggests

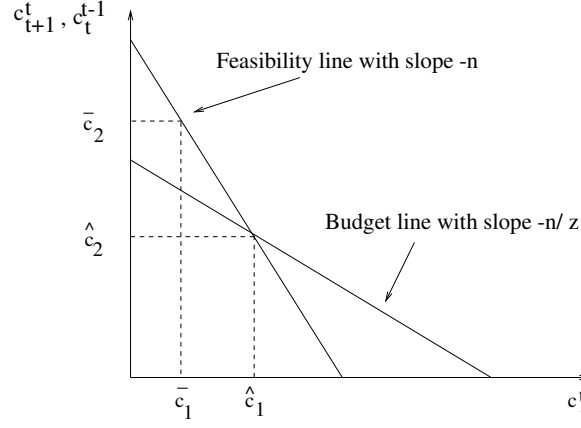


Figure 10.11.1: The feasibility line (10.11.15) and the budget line (10.11.17) when $z > 1$. The consumption allocation in the monetary equilibrium is $(\hat{c}_1, \hat{c}_2)$, and the point that maximizes utility subject to the feasibility line is denoted $(\bar{c}_1, \bar{c}_2)$.

that the quantity of money should not be increased. In particular, if $z \leq 1$, then an optimal $\hat{m}$ equilibrium exists whenever the nonmonetary equilibrium is nonoptimal.

10.11.1. Balasko-Shell criterion for optimality

For the case of constant population, Balasko and Shell (1980) have established a convenient general criterion for testing whether allocations are optimal.¹⁶ Balasko and Shell permit diversity among agents in terms of endowments $[w_t^{th}, w_{t+1}^{th}]$ and utility functions $u^{th}(c_t^{th}, c_{t+1}^{th})$, where w_s^{th} is the time s endowment of an agent named h who is born at t and c_s^{th} is the time s consumption of agent named h born at t . Balasko and Shell assume fixed populations of types h over time. They impose several kinds of technical conditions that serve to rule out possible pathologies. The two main ones are these. First, they assume

¹⁶ Balasko and Shell credit David Cass (1971) with having authored a version of their criterion.

that indifference curves have neither flat parts nor kinks, and they also rule out indifference curves with flat parts or kinks as limits of sequences of indifference curves for given h as $t \rightarrow \infty$. Second, they assume that the aggregate endowments $\sum_h (w_t^{th} + w_t^{t-1,h})$ are uniformly bounded from above and that there exists an $\epsilon > 0$ such that $w_t^{sh} > \epsilon$ for all s, h , and for $t \in \{s, s+1\}$. They consider consumption allocations uniformly bounded away from the axes. With these conditions, Balasko and Shell consider the class of allocations in which all young agents at t share a common marginal rate of substitution $1 + r_t \equiv u_1^{th}(c_t^{th}, c_{t+1}^{th})/u_2^{th}(c_t^{th}, c_{t+1}^{th})$ and in which all of the endowments are consumed. Then Balasko and Shell show that an allocation is Pareto optimal if and only if

$$\sum_{t=0}^{\infty} \prod_{s=0}^t [1 + r_s] = +\infty, \quad (10.11.18)$$

that is, if and only if the infinite sum of t -period gross interest rates, $\prod_{s=0}^t [1 + r_s]$, diverges.

The Balasko-Shell criterion for optimality succinctly summarizes the sense in which low-interest-rate economies are not optimal. We have already encountered repeated examples of the situation that, before an equilibrium with valued currency can exist, the equilibrium without valued currency must be a low-interest-rate economy in just the sense identified by Balasko and Shell's criterion, (10.11.18). Furthermore, by applying the Balasko-Shell criterion, (10.11.18), or generalizations of it that allow for a positive net growth rate of population n , it can be shown that, among equilibria with valued currency, only equilibria with high rates of return on currency are optimal.

10.12. Dynamic inefficiency in a growth model

The starting point of our analysis is a version of the optimal growth example with an infinitely-lived representative household in chapter 3. Our decentralization of the allocation in a competitive equilibrium foreshadows our presentation of growth models in chapters 12 and 13. We use the efficient equilibrium outcomes in models with infinitely-lived agents as a benchmark against which we can compare possibly inefficient equilibrium allocations that entail overaccumulation of capital.

10.12.1. Technology and input prices

The planner in the optimal growth example of chapter 3 solves the following problem:

$$\max_{\{c_t, k_{t+1}, l_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln(c_t)$$

where maximization is subject to

$$\begin{aligned} c_t + k_{t+1} &= A k_t^\alpha l_t^{1-\alpha} \equiv F(k_t, l_t), \\ c_t &\geq 0, \quad k_{t+1} \geq 0, \quad l_t \in [0, 1], \\ &\text{given } k_0 > 0, \end{aligned} \tag{10.12.1}$$

where $A > 0$, $\alpha \in (0, 1)$, and $\beta \in (0, 1)$. We have added labor supply l_t as a choice variable and an input to the production function $F(k_t, l_t)$ in resource constraint (10.12.1). The household is endowed with one unit of time in each period that it can allocate to working, so $l_t \in [0, 1]$. Since the time spent working does not enter the period utility function, $\ln(c_t) \equiv u(c_t)$, the planner will set l_t equal to its highest value. After plugging that optimal value $l_t = 1$ into resource constraint (10.12.1), the planner's optimization problem becomes identical to one formulated in chapter 3.¹⁷

¹⁷ As will become evident, an important feature of our augmented production function in equation (10.12.1) is that the exponents on capital and labor add up to one. This makes the technology have constant returns to scale, $\lambda F(k, l) = F(\lambda k, \lambda l)$ for any $\lambda > 0$. By setting $\lambda = 1/l$, we see that the capital-labor ratio is the determinant of average labor productivity, $F(k, l)/l = F(k/l, 1)$, and, hence, the size of the economy *per se* does not matter. This explains why assuming that the household's time endowment equals one makes the production function,

As discussed in chapter 3, the optimal solution satisfies an Euler equation,

$$1 = \beta \frac{u'(c_{t+1})}{u'(c_t)} F_1(k_{t+1}, 1), \quad (10.12.2)$$

where $F_j(k, l)$ is the partial derivative with respect to the j -th argument and we have invoked the optimal labor supply $l_{t+1} = 1$. When the economy has converged to a stationary value of capital, as given by $k_\infty = (A\beta\alpha)^{\frac{1}{1-\alpha}}$ in chapter 3, the associated constant value of consumption can be deduced from resource constraint (10.12.1). Thus, the Euler equation in a steady state simplifies to

$$F_1(k_\infty, 1) = \frac{1}{\beta}. \quad (10.12.3)$$

A competitive equilibrium in a production economy with infinitely-lived agents is Pareto efficient (see chapter 13). There are several ways to formulate a competitive equilibrium, each with its own versions of optimization problems that confront price-taking households and firms (managers of the production process). Here we adopt a common assumption that households own the capital and rent it to firms at a rental rate r_t , and sell their labor services at a wage rate w_t . Profit maximization by firms implies that the marginal products of capital and labor equal the rental rate and the wage rate, respectively. Thus, in a competitive equilibrium,

$$r_t = F_1(k_t, l_t) = \alpha A \left(\frac{k_t}{l_t} \right)^{-(1-\alpha)} \equiv r(k_t/l_t), \quad (10.12.4a)$$

$$w_t = F_2(k_t, l_t) = (1 - \alpha) A \left(\frac{k_t}{l_t} \right)^\alpha \equiv w(k_t/l_t), \quad (10.12.4b)$$

where equilibrium labor supply is $l_t = 1$.¹⁸

$F(k_t, 1) = A k_t^\alpha$, identical to the one in the growth example of chapter 3 where capital is the sole input; though, note that k_t in the present model is both the capital stock and the capital-labor ratio.

¹⁸ As noted in footnote 17, the production function is homogeneous of degree 1 so, by Euler's theorem, the function equals the sum of its partial derivatives multiplied by their respective variables:

$$F(k, l) = F_1(k, l)k + F_2(k, l)l = rk + wl = \alpha F(k, l) + (1 - \alpha)F(k, l),$$

where the second and third equalities invoke equilibrium conditions (10.12.4). Thus, output is fully distributed as payments to the two production factors. With our Cobb-Douglas production function, those factor shares are constant over time and equal to α for capital and $1 - \alpha$ for labor. For further details, see chapter 13.

In contrast to the efficient competitive equilibria in models with infinitely-lived agents, an equilibrium allocation in a growth model with overlapping generations can fail to be Pareto efficient, a possibility that we explore next.

10.12.2. Agents' optimization and market clearing

The initial old representative agent $i = -1$ in the overlapping generations model is endowed with k_0 units of capital. The sole endowment of all other agents $i \geq 0$ is one unit of time when young that can be devoted to working, while old agents are unable to work. A young agent in period $t \geq 0$ supplies labor l_t^t and purchases goods for current consumption c_t^t and investment that will turn into capital k_{t+1}^t next period (which will finance old-age consumption c_{t+1}^t). The agent solves the following optimization problem:

$$\max_{\{c_t^t, c_{t+1}^t, k_{t+1}^t, l_t^t\}} \ln(c_t^t) + \beta \ln(c_{t+1}^t) \quad (10.12.5)$$

subject to

$$c_t^t + k_{t+1}^t \leq w_t l_t^t, \quad (10.12.6a)$$

$$c_{t+1}^t \leq r_{t+1} k_{t+1}^t, \quad (10.12.6b)$$

$$c_t^t, c_{t+1}^t, k_{t+1}^t \geq 0; l_t^t \in [0, 1].$$

Here we have added a discount factor $\beta > 0$ to earlier preference specification (10.1.1a).¹⁹ Since there is no disutility from working, the agent chooses to set $l_t^t = 1$.

The operation of firms and the characterization of equilibrium factor prices in expression (10.12.4) remain the same as in the model with infinitely-lived agents. The same Euler equation (10.12.2) emerges from the first-order conditions of the present optimization problem (10.12.5):

$$1 = \beta \frac{u'(c_{t+1}^t)}{u'(c_t^t)} r_{t+1} = \beta \frac{u'(c_{t+1}^t)}{u'(c_t^t)} F_1(k_{t+1}, 1), \quad (10.12.7)$$

¹⁹ In the case of infinitely-lived agents in the preceding subsection, we imposed the restriction that $\beta \in (0, 1)$ to ensure that an agent's lifetime utility is bounded. That is no longer necessary when an agent's life length is finite. Parameterizations with $\beta \geq 1$ are common in applied work using overlapping generations models; a larger discount factor generates larger savings, which helps to match aggregate observations of high capital-output ratios.

where the second equality follows from equilibrium condition (10.12.4a) and we have invoked equilibrium labor supply $l_{t+1} = 1$. But Euler equation (10.12.7) now pertains to a life-cycle optimization problem where the agent is only concerned about the two periods t and $t + 1$.

Under our assumption that $u(c) = \ln(c)$, the saving function is $s(w_t, r_{t+1}) = w_t - c_t^t = w_t - \frac{1}{1+\beta}w_t = \frac{\beta}{1+\beta}w_t$.²⁰ In an equilibrium, saving of the representative agent $t \geq 0$ equals next period's capital stock k_{t+1} :

$$k_{t+1} = s(w_t, r_{t+1}) = \frac{\beta}{1+\beta}w_t = \frac{\beta}{1+\beta}(1-\alpha)Ak_t^\alpha, \quad (10.12.8)$$

where the third equality follows from equilibrium condition (10.12.4b) and we have invoked equilibrium labor supply $l_t = 1$. The law of motion for capital in equation (10.12.8) converges to a steady state in which capital equals $\hat{k} = [\frac{\beta}{1+\beta}(1-\alpha)A]^{\frac{1}{1-\alpha}}$.

10.12.3. Dynamic inefficiency and Pareto improvements

In appendix A, we generalize our model by allowing for population growth, $N_t = nN_{t-1}$ with $n > 0$, and by introducing a capital depreciation rate $\delta \in (0, 1]$ that can be less than the present assumption of 100% depreciation. With those extra features, the following proposition delineates when a stationary equilibrium fails to be Pareto optimal.

PROPOSITION: Suppose a competitive equilibrium converges to a steady state with capital-labor ratio k^* satisfying

$$F_1(k^*, 1) < \delta + n - 1,$$

then the equilibrium allocation is not Pareto optimal.

In our setting with $\delta = n = 1$, the condition for inefficiency translates into $F_1(k^*, 1) < 1$, which implies that $k^* > [\alpha A]^{\frac{1}{1-\alpha}} \equiv \bar{k}$. We can compute a set of

²⁰ With logarithmic preferences, the intertemporal elasticity of substitution is equal to one, so when the rate of return on savings rises, the incentive to consume later (substitution effect) is exactly offset by the increased wealth that allows for higher consumption today (income effect). Thus, variations in the rental rate on capital r_{t+1} do not affect optimal savings.

parameter values that make the steady-state value $\hat{k}$, as derived above, exceed $\bar{k}$:

$$\hat{k} > \bar{k} \implies \frac{\beta}{1+\beta} \frac{1-\alpha}{\alpha} > 1, \quad (10.12.9)$$

which requires that $(1-\alpha) > \alpha$ since $\beta/(1+\beta) < 1$. As one might have anticipated, parameterizations that would lead young agents to choose relatively high savings can depress rates of return on capital so much that there is “dynamic inefficiency” in the sense of an overaccumulation of capital. This outcome is more “likely” the greater the discount factor β is, i.e., when agents place a high weight on the utility of future consumption, and the larger the labor share of output $(1-\alpha)$ is relative to the capital share (α) , so that young agents receive relatively high labor earnings (see footnote 18 about factor shares).

To prove the condition for inefficiency, we show by example a possible Pareto improvement that moves the economy from an inefficient steady state to a Pareto optimal steady state in just one transition period. A government policy that implements such a Pareto improvement in a competitive equilibrium has two parts: (i) a one-time lump-sum tax and transfer scheme in the transition period, and (ii) the issue and eternal rollover of unbacked government debt.

Our welfare analysis starts with the resource constraint (and goods market clearing condition) in period $t \geq 0$:

$$c_t^t + c_t^{t-1} + k_{t+1} = F(k_t, 1), \quad (10.12.10)$$

which, at a stationary allocation with capital k , can be rearranged to read

$$c_1 + c_2 = F(k, 1) - k \equiv C(k), \quad (10.12.11)$$

where c_1 and c_2 are the consumption of a young and an old agent, respectively. Figure 10.12.1 depicts components of expression (10.12.11) as functions of a stationary value of k . Hence, the stationary aggregate consumption $C(k)$ on the solid line is the difference between the two dashed lines: aggregate output $F(k, 1)$ minus the stationary value of capital k . Since the slope of the latter dashed diagonal line is one, consumption $C(k)$ attains a maximum when the marginal product of capital has fallen to one, which occurs at $\bar{k}$ as computed above. The maximum sustainable level of consumption $C(\bar{k})$ is known as the Golden Rule of capital accumulation.²¹

²¹ Evidently, if the initial capital endowment of the infinitely-lived household in subsection 10.12.1 is $k_0 = \bar{k}$, i.e., the Golden-Rule level of capital, an efficient allocation would

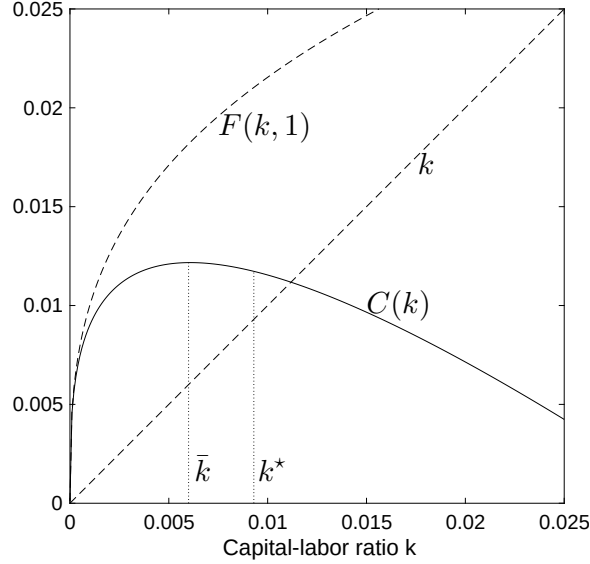


Figure 10.12.1: Output $F(k, 1)$ and consumption $C(k)$ as functions of a stationary capital-labor ratio k . The Golden-Rule level of k is $\bar{k}$, i.e., $C'(\bar{k}) = 0$, and k^* is an example of a dynamically inefficient steady state.

For the sake of argument, consider a steady-state equilibrium with capital k^* that exceeds $\bar{k}$, i.e., capital k^* is on the downward-sloping side of $C(k)$ as illustrated in Figure 10.12.1. Starting in some period T , the following change in capital expands aggregate consumption in all periods $t \geq T$ and, hence, it would be feasible to increase consumption of all currently living and all future agents. Specifically, the planner in period T permanently lowers next period's and all future periods' capital from k^* to $\bar{k}$. Thus, the resource constraint in period T becomes

$$c_T^T + c_T^{T-1} + \bar{k} = F(k^*, 1) = C(k^*) + k^*, \quad (10.12.12)$$

which can be rearranged to read

$$c_T^T + c_T^{T-1} = C(k^*) + k^* - \bar{k} > C(k^*). \quad (10.12.13)$$

gradually reduce capital until it reaches k_∞ , the value prescribed by the steady-state Euler equation (10.12.3). The utility of temporarily higher consumption outweighs the ultimate and permanently lower level of consumption.

Not surprisingly, aggregate consumption in period T is increased by the reduction in capital investment, i.e., by $k^* - \bar{k} > 0$. In all future periods $t > T$, aggregate consumption is also increased since $C(\bar{k}) > C(k^*)$ in Figure 10.12.1.

This change in capital can be attained in a competitive equilibrium with the following government policy.²² In the transition period T , an old agent receives a lump-sum transfer $\tau_T^{T-1} < 0$ as computed below, while a young agent pays a lump-sum tax $\tau_T^T > 0$ so that the disposable income becomes the same as the labor earnings of all future young agents whose wages will be suppressed under the targeted lower future capital $\bar{k}$,

$$\tau_T^T = w(k^*) - w(\bar{k}) > 0. \quad (10.12.14a)$$

The government policy so far would simply set the current young and future agents back as if the economy had an imaginary capital equal to $\bar{k}$ already at the beginning of period T and then, according to equilibrium law of motion (10.12.8), capital would eventually revert to its higher steady-state value k^* and rate of return $r(k^*) < 1$. So in order to freeze future capital at $k_{T+1} = \bar{k}$ and rate of return $r(\bar{k}) = 1$, the government in period T issues one-period bonds at a sale price equal to the bonds' face value, i.e., the debt policy sets a floor for the equilibrium gross interest rate, $R_T \geq 1$. Specifically, if an equilibrium without government bonds has an interest rate $R_T > 1$, there would be no demand for those bonds; but if sold in an equilibrium, a no-arbitrage argument ensures that not only bonds but also capital would earn the same gross rate of return equal to one. Any issued bonds are rolled over forever thereafter. Given our assumption $k^* > \bar{k}$, the strictly positive quantity of bonds required to support the interest rate $R_T = 1$ can be deduced from the market clearing condition for savings in period T ,

$$s(w(k^*) - \tau_T^T, r(\bar{k})) = s(w(\bar{k}), r(\bar{k})) = \bar{k} + \frac{B_{T+1}}{R_T}, \quad (10.12.14b)$$

where $r(\bar{k}) = 1 = R_T$. The government policy must also be budget feasible:

$$\tau_T^{T-1} = - \left(\tau_T^T + \frac{B_{T+1}}{R_T} \right), \quad (10.12.14c)$$

$$\frac{B_{t+1}}{R_t} = B_t, \quad \forall t \geq T+1, \quad (10.12.14d)$$

²² We assume that the policy comes as a surprise to the agents when implemented in period T . If the policy had been known in advance in period $T-1$, young agents in period $T-1$ would have changed their savings because of a lump-sum transfer they were about to receive in period T and, hence, it would have altered the capital stock k_T .

where equation (10.12.14c) is the budget constraint in period T (that residually determines τ_T^{T-1}) and equations (10.12.14d) are the future budget constraints, all of which are evaluated at the equilibrium interest rates $R_t = 1$ for all $t \geq T$. Evidently, the quantity of bonds stays constant over time and can be computed from market clearing condition (10.12.14b),

$$\bar{B} = s(w(\bar{k}), 1) - \bar{k} = \frac{\beta}{1+\beta} w(\bar{k}) - \bar{k} > 0, \quad (10.12.15)$$

where the strict inequality is implied by $\bar{k}$ being less than the original steady-state value k^* , i.e., parameter restriction (10.12.9) holds.²³

This government policy results in a Pareto improvement where all currently living and all future agents are strictly better off. The old agent at time T is trivially better off when receiving additional income from the lump-sum transfer that equals the sum of the lump-sum tax paid by the young agent in period T and the funds raised by the government from the first issue of one-period bonds in period T . All young agents $t \geq T$ are better off in spite of receiving a smaller income $w(\bar{k})$ that is less than their wage earnings $w(k^*)$ in the original steady state. First, the fact that $C(\bar{k}) > C(k^*)$ implies that the sum of young-age and old-age consumption is greater in the new steady state as compared to the original steady state. Second, because they derive all their income in young age and, hence, are savers, they benefit from the higher interest rate in the new steady state as they attain a more favorable intertemporal consumption profile – consuming less in young age, $\bar{c}_1$, and more in old age, $\bar{c}_2$, as compared to the original steady state, (c_1^*, c_2^*) :

$$\bar{c}_1 = \frac{1}{1+\beta} w(\bar{k}) < \frac{1}{1+\beta} w(k^*) = c_1^*, \quad (10.12.16a)$$

where the strict inequality is directly implied by $\bar{k} < k^*$ and that the wage is equal to the marginal product of labor in equation (10.12.4b);

$$\bar{c}_2 = s(w(\bar{k}), 1) > r(k^*) s(w(k^*), r(k^*)) = c_2^*, \quad (10.12.16b)$$

²³ To prove the strict inequality in expression (10.12.15), substitute equation (10.12.4b) and $\bar{k} = [\alpha A]^{\frac{1}{1-\alpha}}$ into that expression to obtain

$$\bar{B} = \frac{\beta}{1+\beta} (1-\alpha) A [\alpha A]^{\frac{\alpha}{1-\alpha}} - [\alpha A]^{\frac{1}{1-\alpha}} = [\alpha A]^{\frac{1}{1-\alpha}} \left[\frac{\beta}{1+\beta} \frac{1-\alpha}{\alpha} - 1 \right] > 0,$$

where the strict inequality follows from parameter restriction (10.12.9).

where the strict inequality follows from parameter restriction (10.12.9).²⁴

10.12.4. Money and the real bills doctrine

In accordance with subsection 10.6.1, we can transform the equilibrium under the government policy with unbacked debt in the preceding subsection into an equilibrium with valued fiat money. The one-time lump-sum tax and transfer scheme in period T is the same but instead of issuing unbacked debt to be rolled over forever, the government once and for all issues a quantity of fiat money $\bar{M}$ in period T . This money is immediately handed over to the current old agent. There is now a single government budget constraint,

$$\tau_T^{T-1} = - \left(\tau_T^T + \frac{\bar{M}}{p_T} \right). \quad (10.12.17)$$

After period T , the government takes no further actions.

Following the steps in subsection 10.6.1, to arrive at an equilibrium with fiat money, pick some arbitrary nominal quantity of money $\bar{M}$ and solve for p_T from $\bar{M}/p_T = B_{T+1}/R_T$. All other prices, $\{p_t\}_{t>T}$, are then recursively set according to $p_{t-1}/p_t = R_{t-1}$ for $t > T$. Since the gross rate of return equals one in the equilibrium with unbacked debt, the price level stays constant in the monetary equilibrium and equals

$$p_t = \bar{p} = \frac{\bar{M}}{\bar{B}}, \quad \forall t \geq T, \quad (10.12.18)$$

where the time-invariant debt level $\bar{B}$ in the earlier equilibrium is given by (10.12.15). The new market clearing condition for savings becomes

$$s(w(\bar{k}), 1) = \bar{k} + \frac{\bar{M}}{\bar{p}}. \quad (10.12.19)$$

²⁴ After substituting savings function $s(w, r) = \frac{\beta}{1+\beta}w$ and factor-price expressions (10.12.4) into expression (10.12.16b) and invoking $\bar{k} = [\alpha A]^{\frac{1}{1-\alpha}}$ and $k^* = \hat{k} = [\frac{\beta}{1+\beta}(1-\alpha)A]^{\frac{1}{1-\alpha}}$, the resulting inequality can be simplified and rearranged to read

$$\left[\frac{\beta}{1+\beta} \frac{1-\alpha}{\alpha} \right]^{1-\frac{\alpha}{1-\alpha}} > 1,$$

where the quantity inside square brackets is greater than one by parameter restriction (10.12.9); a restriction that also implies that $\alpha < (1-\alpha)$ and, hence, the quantity greater than one in square brackets is raised to a strictly positive power. Thus, the strict inequality here holds and, hence, also holds for expression (10.12.16b).

Next, to study the real bills doctrine, we allow the government to issue additional money after period T . We show that so long as the additional money is used to acquire capital, any expansions of the money supply will leave the price level and equilibrium allocation unchanged. Let $k_{t+1}^g \in [0, \bar{k}]$ be the capital held by the government between periods t and $t+1$. Under an assumption that these open market operations leave the price level and allocation unchanged, the government budget constraints (and constraints on open market operations) become

$$\frac{M_t - M_{t-1}}{\bar{p}} + r(\bar{k})k_t^g = k_{t+1}^g, \quad \forall t \geq T+1, \quad (10.12.20)$$

where $M_T = \bar{M}$ and $k_{T+1}^g = 0$. Equation (10.12.20) states that the government finances capital holding k_{t+1}^g in two ways: first, by injecting new money in the amount $M_t - M_{t-1}$, and second, by rolling over the proceeds $r(\bar{k})k_t^g$ from last period's capital holding where $r(\bar{k}) = 1$. Furthermore, the open market operations change the market clearing condition (10.12.19) for private agents' savings to become

$$s(w(\bar{k}), 1) = \bar{k} - k_{t+1}^g + \frac{M_t}{\bar{p}}, \quad \forall t \geq T+1, \quad (10.12.21)$$

where the net of the first two terms on the right side is the capital held by private agents and the third term is the aggregate real money supply.

Effects of open market operations like this have concerned monetary economists for centuries under several guises.²⁵ We state the following proposition:

IRRELEVANCE OF OPEN MARKET OPERATIONS: Sequences $\{M_t \geq \bar{M}, k_{t+1}^g \in [0, \bar{k}]\}_{t=T+1}^\infty$ that satisfy the constraint (10.12.20) on open market operations are associated with the same equilibrium allocation $\{\bar{c}_1, \bar{c}_2, \bar{k}\}$ as derived in subsection 10.12.3 and price level $\bar{p}$ in equation (10.12.18).

PROOF: We can rearrange and manipulate the government budget constraint (10.12.20) for period $t = T+j$ as follows

$$\begin{aligned} \frac{M_{T+j}}{\bar{p}} &= k_{T+j+1}^g - k_{T+j}^g + \frac{M_{T+j-1}}{\bar{p}} \\ &= \sum_{i=0}^{j-1} (k_{T+j+1-i}^g - k_{T+j-i}^g) + \frac{M_T}{\bar{p}} = k_{T+j+1}^g + \frac{\bar{M}}{\bar{p}}, \end{aligned} \quad (10.12.22)$$

²⁵ One issue concerned the effects on the price level of allowing banks to issue private bank notes. Nothing in our model requires that the notes M_t be issued by the government. We can also think of them as being issued by a private bank.

where the first equality invokes $r(\bar{k}) = 1$, the second equality solves the equation backward to period $T + 1$, and the third equality invokes initial conditions $k_{T+1}^g = 0$ and $M_T = \bar{M}$. Substituting (10.12.22) into market clearing condition (10.12.21) for period $t = T + j$ gives

$$s(w(\bar{k}), 1) = \bar{k} - k_{T+j+1}^g + \frac{M_{T+j}}{\bar{p}} = \bar{k} + \frac{\bar{M}}{\bar{p}}, \quad (10.12.23)$$

which is the same as market clearing condition (10.12.19) in the economy with no open market operations. Evidently, it does not matter if the government or the private sector owns the capital when the additional money issued for the government's acquisition has the same rate of return and is fully backed by the capital. Specifically, when the government stops rolling over its capital holding, the constraint (10.12.20) on open market operations prescribes that the corresponding quantity of money is redeemed by the government and de facto returns the ownership of capital to the private sector. Thus, the invisible hand ensures that the sum of private sector and government holdings of capital equals $\bar{k}$. ■

This proposition captures the spirit of Adam Smith's real bills doctrine, which states that if the government issues bank notes to purchase safe evidences of private indebtedness, it is not inflationary. The quantity $M_t - \bar{M}$ is sometimes called "inside money" that is 100% backed by evidences of private indebtedness while $\bar{M}$ is called "outside money" that is "unbacked". Sargent and Wallace (1982) analyze settings in which the money market is initially separated from the credit market by some legal restrictions that inhibit intermediation. Then open market operations are no longer irrelevant because they can partially undo the legal restrictions. Sargent and Wallace show how those legal restrictions can help stabilize the price level at a cost in terms of economic efficiency. Kahn and Roberds (1998) extend the Sargent and Wallace model to study issues about regulating electronic payments systems.

10.13. Concluding remarks

The overlapping generations model is a workhorse in analyses of public finance, welfare economics, and demographics. Diamond (1965) studied some fiscal policy issues within a version of the model with a neoclassical production. He showed that, depending on preference and productivity parameters, equilibria of the model can have too much capital, and that such capital overaccumulation can be corrected by having the government issue and perpetually roll over unbacked debt.²⁶ Auerbach and Kotlikoff (1987) formulated a long-lived overlapping generations model with capital, labor, production, and various kinds of taxes. They used the model to study a host of fiscal issues. Rios-Rull (1994a) used a calibrated overlapping generations growth model to examine the quantitative importance of market incompleteness for insuring against aggregate risk. See Attanasio (2000) for a review of theories and evidence about consumption within life-cycle models.

Several authors in a 1980 volume edited by John Kareken and Neil Wallace argued through example that the overlapping generations model is useful for analyzing a variety of issues in monetary economics. We refer to that volume, McCandless and Wallace (1992), Champ and Freeman (1994), Brock (1990), and Sargent (1987b) for a variety of applications of the overlapping generations model to issues in monetary economics.

A. Dynamic inefficiency in a generalized growth model

This appendix generalizes the growth model with overlapping generations in section 10.12 by allowing for population growth, $N_t = nN_{t-1}$ with $n > 0$, and for any rate of depreciation of capital, $\delta \in (0, 1]$. By so relaxing our earlier assumptions $n = \delta = 1$, we can prove the proposition on suboptimal steady states in subsection 10.12.3, i.e., a steady state with a capital-labor ratio k^* satisfying $F_1(k^*, 1) < \delta + n - 1$ is Pareto suboptimal.

The earlier resource constraint (10.12.10) in period $t \geq 0$ is replaced by

$$N_t c_t^t + N_{t-1} c_t^{t-1} + K_{t+1} = F(K_t, N_t) + (1 - \delta)K_t, \quad (10.A.1)$$

²⁶ Abel, Mankiw, Summers, and Zeckhauser (1989) propose an empirical test of whether there is capital overaccumulation in the U.S. economy, and conclude that there is not.

where K_i denotes the aggregate capital stock accumulated by the end of period $i - 1$ to be used in production in period i (and in future periods after deducting for depreciation). Hence, the inputs in the production function are capital K_t and labor supplied by the current young generation of size N_t . Dividing both sides of resource constraint (10.A.1) by N_t yields

$$c_t^t + n^{-1}c_t^{t-1} + nk_{t+1} = F(k_t, 1) + (1 - \delta)k_t, \quad (10.A.2)$$

where $k_i \equiv K_i/N_i$ is the capital-labor ratio.²⁷ At a stationary allocation with capital-labor ratio k , equation (10.A.2) can be rearranged to read

$$c_1 + n^{-1}c_2 = F(k, 1) - (\delta + n - 1)k \equiv C(k), \quad (10.A.3)$$

where c_1 and c_2 are the consumption of a young and an old agent, respectively. We can now reinterpret Figure 10.12.1 to represent components of expression (10.A.3) as functions of a stationary capital-labor ratio k . While the upper dashed line $F(k, 1)$ would be the same as before, the slope of the straight dashed line would now be $(\delta + n - 1)$. The solid line is the difference between the two dashed lines, which represents aggregate consumption where the factor n^{-1} accounts for different relative cohort sizes of old versus young agents.

As before, a Pareto suboptimal steady state has a capital-labor ratio k^* that is on the downward-sloping side of $C(k)$, which here means that $F_1(k^*, 1) < \delta + n - 1$. Following the steps in subsection 10.12.3, we describe a Pareto improvement that moves the economy from such an inefficient steady state to a Pareto optimal steady state in one transition period. Specifically, the planner in period T permanently lowers next period's and all future periods' capital-labor ratio from k^* to $\bar{k}$, where $\bar{k}$ is the capital-labor ratio at which $C'(\bar{k}) = 0$. Thus, the resource constraint in period T becomes

$$c_T^T + n^{-1}c_T^{T-1} + n\bar{k} = F(k^*, 1) + (1 - \delta)k^* = C(k^*) + nk^*, \quad (10.A.4)$$

which can be rearranged to read

$$c_T^T + n^{-1}c_T^{T-1} = C(k^*) + n(k^* - \bar{k}) > C(k^*), \quad (10.A.5)$$

²⁷ As shown in footnote 17, $F(K, N) = NF(K/N, 1)$. Note that our earlier interchangeable use of lowercase letter k_t to denote both the capital stock and the capital-labor ratio in section 10.12 was dependent on the representative young agent's time endowment being equal to one. In our present setting, we must distinguish between the lowercase letter k_t denoting the capital-labor ratio and the uppercase letter K_t denoting the aggregate capital stock.

where the strict inequality follows from $k^* > \bar{k}$. Aggregate consumption is increased in period T according to expression (10.A.5), as well as in all future periods $t > T$ since $C(\bar{k}) > C(k^*)$ in our reinterpretation of Figure 10.12.1.

The condition for a steady state to be Pareto suboptimal, $F_1(k^*, 1) < \delta + n - 1$, is about the marginal product of capital $F_1(k^*, 1)$ being too low.²⁸ First, if $F_1(k^*, 1)$ is less than the rate of depreciation δ , it would mean that the marginal unit of capital does not increase output enough to replace its own depreciation. Second, the cutoff for suboptimality is further raised if the population grows, $n > 1$. This is also a kind of required replacement as seen from the planner's perspective; namely, the marginal unit of capital should not only exceed its own depreciation rate but also yield an additional $n - 1$ so as to create the extra output needed to maintain that higher capital-labor ratio in the face of a growing labor force. If the marginal product of capital $F_1(k^*, 1)$ falls short of these requirements, it is efficient to reduce capital investments and instead set up a system of intratemporal transfers of goods from young to old agents, which can be done with unbacked government debt or fiat money.

Exercises

Exercise 10.1 At each date $t \geq 1$, an economy consists of overlapping generations of a constant number N of two-period-lived agents. Young agents born in t have preferences over consumption streams of a single good that are ordered by $u(c_t^i) + u(c_{t+1}^i)$, where $u(c) = c^{1-\gamma}/(1-\gamma)$, and where c_t^i is the consumption of an agent born at i in time t . It is understood that $\gamma > 0$, and that when $\gamma = 1$, $u(c) = \ln c$. Each young agent born at $t \geq 1$ has identical preferences and endowment pattern (w_1, w_2) , where w_1 is the endowment when young and w_2 is the endowment when old. Assume $0 < w_2 < w_1$. In addition, there are some initial old agents at time 1 who are endowed with w_2 of the time 1

²⁸ Under our assumption of constant returns to scale, $F_1(k, 1)$ is the marginal product of capital for any pair (K, N) whose ratio satisfies $K/N = k$;

$$\frac{\partial F(K, N)}{\partial K} = \frac{\partial NF\left(\frac{K}{N}, 1\right)}{\partial K} = N \frac{1}{N} \frac{\partial F(k, 1)}{\partial k} = F_1(k, 1).$$